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Review

Intelligent algorithms and control strategies for battery management system in electric vehicles: Progress, challenges and future outlook

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ABSTRACT

Globally, the research on battery technology in electric vehicle applications is advancing tremendously to address the carbon emissions and global warming issues. The effectiveness of electric vehicles depends on the accurate assessment of key parameters as well as proper functionality and diagnosis of the battery storage system. However, poor monitoring and safety strategies of the battery storage system can lead to critical issues such as battery overcharging, over-discharging, overheating, cell unbalancing, thermal runaway, and fire hazards. To address these concerns, an effective battery management system plays a crucial role in enhancing battery performance including precise monitoring, charging-discharging control, heat management, battery safety, and protection. The goal of this paper is to deliver a comprehensive review of different intelligent approaches and control schemes of the battery management system in electric vehicle applications. In line with that, the review evaluates the intelligent algorithms in battery state estimation concerning their features, structure, configuration, accuracy, advantages, and disadvantages. Moreover, the review explores the various controllers in battery heating, cooling, equalization, and protection highlighting categories, characteristics, targets, achievements, benefits, and shortcomings. The key issues and challenges in terms of computation complexity, execution problems along with various internal and external factors are identified. Finally, future opportunities and directions are delivered to design an efficient intelligent algorithm and controller toward the development of an advanced battery management system for future sustainable electric vehicle applications.

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1. Introduction

Nowadays, the automotive industry has made great strides due to the various technological advancements toward the improvement of both passengers and pedestrians' safety (Gohoungodji et al., 2020). However, the greater number of vehicles on the roads is responsible to increase the pollution levels dramatically in urban environments (Guo et al., 2020). The European Union has reported that the transport sector is accountable for approximately 27% of the total carbon dioxide (CO_2) emissions, while the vehicle transport is responsible for over 70% of the emissions (EEA, 2017). To tackle these problems, electric vehicles (EVs) have received massive attention and popularity around the world due to their capability to decrease environmental pollution, reserve fossil fuels, minimize carbon emission and global warming issues (Lin et al., 2020). EVs are the promising alternative to diesel-based vehicles not only at emissions levels but also in terms of simplicity, reliability, comfort and efficiency (Zheng et al., 2020). However, wide adoption of EV requires proper functionalities and diagnosis of the battery storage system (BSS) in terms of battery cell monitoring, charge-discharge control, cell balancing, heat management, and power management control (Kumar and Alok, 2020).

Currently, lithium-ion batteries are dominant in the EV battery market due to their high power and energy density, high voltage, extended life cycles and low self-discharge rates (Nikolian et al., 2016). Nevertheless, lithium batteries are sensitive to aging and temperature; thus, special focus is required on their working environments to avoid any physical damage, aging, and thermal runaways (Aaldering et al., 2019) (Xiong et al., 2020b). The battery management system (BMS) in EV operation is necessary to monitor battery current, voltage, temperature; examine battery charge, energy, health, equalize the voltage among cells, control temperature, and identify the fault (Lin et al., 2019). An efficient BMS has the following key responsibilities: (i) estimates and evaluates the battery states accurately including state of charge (SOC), state of energy (SOE), state of health (SOH) and remaining useful life (RUL), (ii) controls the battery temperatures within the safe limit, (iii) operates fault diagnosis, fault prognosis, and fault handling and (iv) balances the voltage, charge, and capacity among battery cells (Hussain et al., 2019). A schematic diagram of an efficient BMS for EV operation is shown in Fig. 1.

BMS in EVs includes many circuits, components, power electronics, sensors, actuators, diode, capacitor, inductor, transformer, switching, converter, and safety equipment which are controlled by many algorithms, models, and control signals (Lelie et al., 2018). A great deal of research has been performed on developing the appropriate algorithms in BMS (Zhang et al., 2018b). The most common approaches employed in BMS include model-based methods and intelligent methods (Lipu et al., 2020a). The intelligent algorithms are suitable for lithium-ion batteries to address complex, dynamic, and nonlinear characteristics (Zhao et al., 2020).

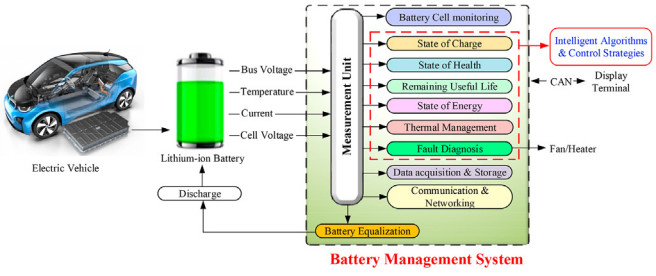


Fig. 1. Schematic diagram of BMS for EV applications.

Besides, intelligent algorithms demonstrate enhanced learning capability, fast convergence speed, improved generalization and high accuracy (Xiong et al., 2018b). Moreover, intelligent algorithms do not require explicit information and knowledge about battery internal characteristics, chemical reactions and working principles (Z. Huang et al., 2019b). Additionally, intelligent algorithms can operate without an added filter, mathematical model, and can update the network parameters by self-learning algorithm which is ideal for battery state estimation (Tian et al., 2020). In line with that, the various optimization techniques are integrated with intelligent algorithms in BMS to improve accuracy, efficiency, adaptability, and robustness (He et al., 2020; Merhy et al., 2020; Tran et al., 2020). An efficient controller of BMS ensures battery safety as well as protects the battery pack from hazard conditions (Ringbeck et al., 2020). Besides, the controller maintains the battery cooling and heating temperature within a safe limit (Hannan et al., 2019). Moreover, the controller in BMS helps to equalize the imbalance of battery packs (Zun et al., 2020).

To date, many BMS related articles have been reported in the literature. Shen and Gao (2019) conducted a survey on model-based BMS highlighting state estimation, temperature control, energy equalization and fault diagnosis. Although the authors listed a large pool of articles related to model-based approaches, they did not provide the classification of methods and outline the intelligent algorithms and control strategies in BMS. Lu et al. (2013) focused on the key issues of BMS for lithium-ion batteries in EV applications. The authors examined the methods of SOC, SOH, battery equalization and faults. However, the explanation was limited to only a few intelligent approaches. Lelie et al. (2018) reviewed the hardware features of BMS in EVs concerning battery pack topology and associated safety, reliability and complexity. Nonetheless, the authors did not cover intelligent methods and control techniques of BMS. Balasingam et al. (2020) explored the various parameters of BMS including SOC, SOH, RUL, power fade, capacity, impedance and optimal charging based on model-based approaches. Nevertheless, intelligent algorithms and control schemes were not investigated in detail. Wang et al. (2020b) conducted a systematic review on BMS highlighting SOC, SOH and RUL estimation as well as battery modeling including data-driven models, electrochemical models and equivalent circuit models. However, the authors did not outline the controller scheme in thermal management, fault diagnosis and battery equalization. Chaturvedi et al. (2010) developed the framework for lithium-ion battery models. Nonetheless, the authors did not discuss the state estimation algorithms and the role of controller schemes in BMS. Xiong (2020) provided a detailed description of model-based SOC, SOH and state of power (SOP) estimation of BMS. Nevertheless, the intelligent algorithms and controller schemes in BMS were not reported. Xiong et al. (2020a) discussed the lithium-ion battery safety protection concerning diagnosing and prognosticating short circuit in BMS. However, battery state estimation, thermal management, battery equalization were not studied. Although Hussain et al. (2019) and Xiong et al., (2018b) presented a detailed classification of battery SOC and SOH in BMS, respectively, but fault diagnosis, temperature control, charge equalization strategies in BMS were not covered. Hannan et al. (2018a) carried out a comprehensive study on the fundamental, structure and performance of lithium-ion battery as well as state estimation, temperature and heat management, fault diagnosis, protection and equalization. However, the intelligent methods and control techniques in BMS were not described extensively. Xing et al. (2011) explained the concerns of BMS in EVs along with battery modeling, state evaluation and cell balancing. Nevertheless, the authors did not investigate various intelligent approaches and control strategies in depth.

To bridge the existing research limitations, this survey unveils new contributions with a detailed explanation of recent intelligent algorithms and control strategies for BMS in EVs. The main findings of this review are summarized below.

- This review comprehensively examines the various intelligent approaches toward SOC, SOE, SOH and RUL estimation in BMS. The intelligent algorithms are classified according to feed-forward algorithms, time-series based learning, hybrid optimization algorithms, and statistical algorithms. In line with that, their structure, configuration, features, estimated error, benefits, weaknesses and research gaps are provided.
- The role of various controllers in the battery pack equalization, diagnosis of faults and thermal management are outlined highlighting their types, characteristics, target, contributions, strength, and weakness.
- The various existing issues and challenges with respect to algorithm complexity, implementation issues as well as data variety, data integrity, structure, uncertainties are discussed rigorously.
- Finally, the review offers useful suggestions for future opportunities and developments of BMS.

The rest of the paper is structured into six sections. Section 2 explains the reviewing methodology. Section 3 covers the comprehensive exploration of intelligent algorithms in battery state estimation. Besides, the various controller schemes in battery equalization, fault diagnosis and thermal management are outlined. The section 4 narrates the existing research gaps, key issues and challenges. The prospects and recommendations for BMS are delivered in section 5. The concluding remarks are depicted in section 6.

2. Reviewing methodology

This review process was performed based on content analysis. The exploration of the relevant literature was carried out using the Scopus databases. The proper references were collected and cited accordingly from Google Scholar, Scopus and Web of Science platforms. The related articles are searched using the important keywords within the scope such as *battery management system, lithium-ion batteries, electric vehicle, state estimation, thermal management, fault diagnosis, battery equalization.* The authors have found several journals and conference papers: however, appropriate articles are chosen by assessing the paper title, abstract, outlines and contributions. After that, the list of articles was finalized using journal impact factor, citation and review process.

The findings of this reviewing approach were structured into four parts. Firstly, the intelligent approaches in battery state estimation were studied comprehensively. Secondly, the role of controllers in battery equalization, fault diagnosis and thermal management were reviewed. Thirdly, the review highlighted the key limitations and challenges of different intelligent algorithms and controllers of BMS. Fourthly, the review suggested some selective proposals for the future advancement of BMS in EVs. The schematic diagram of the reviewing methodology is presented in Fig. 2. The method of this review was split into two stages, as illustrated below.

2.1. Selection process

- A total of 778 articles was identified after the primary screening.
- The second screening was performed using the proper keywords, title, abstract, subject, contributions and therefore 423 articles were selected.

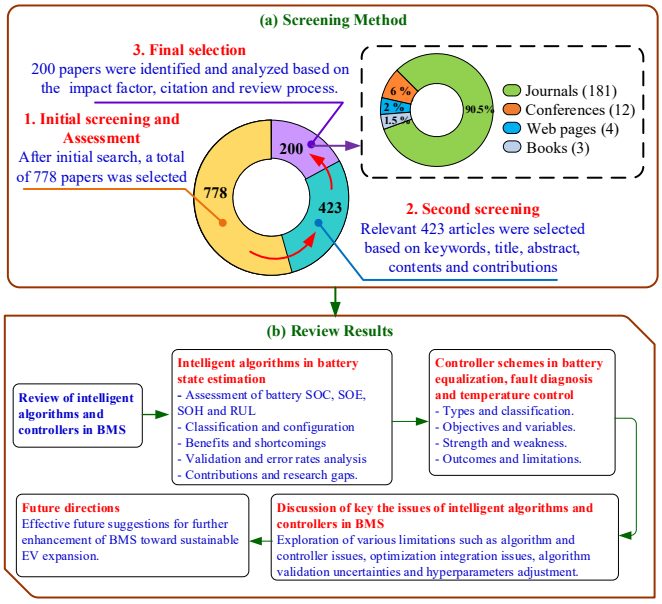


Fig. 2. Schematic diagram of the reviewing methodology.

- The third screening was carried out based on the impact factor, the number of citations and the review process.
- Finally, a total of 200 articles published in prominent scientific conference proceedings and notable journals were identified for analysis and discussion.

2.2. Results of the review

- The value of the intelligent algorithms toward accurate battery state estimation including SOC, SOH, RUL and SOH was broadly surveyed.
- Various controller schemes in battery equalization, fault diagnosis and thermal management were explicitly reviewed.
- The key issues and challenges of intelligent algorithms and controllers for BMS were discussed.
- Selective future directions for further improvement of BMS to elevate accuracy, adaptability and robustness were provided.

3. Progress of intelligent algorithms and control schemes for BMS

This section provides a comprehensive exploration of BMS operation including state estimation, battery equalization, fault diagnosis and thermal management. The recent progress of intelligent algorithms and control strategies are highlighted to explain the BMS operation.

3.1. Battery state estimation in BMS

The accurate evaluation of battery states enhances battery aging performance, extends battery life, and confirms a secure and reliable driving of EV. Commonly, battery state estimation approaches in BMS is classified into two groups; model-based and intelligent methods (Xiong et al., 2018a). The model-based state estimation approaches require detailed domain knowledge, a deeper understanding of battery and a moderate amount of data to develop robust rules and mathematical model (Shrivastava et al., 2019). Moreover, the comprehensive theoretical knowledge related to physics and chemistry such as anode, cathode material features, electrochemical reaction, polarization, solid-phase diffusion containing heavy computation and functional relationships are needed to be studied (D. Huang et al., 2019a). Furthermore, extensive research, domain experts and a considerable amount of duration are required to conduct the experimental tests (Lai et al., 2020). The model-based SOC estimation methods have shorter offline operational time and obtain reasonable solutions if the parameters of battery model are assessed accurately (Bi and Choe, 2020). However, they have a lengthy and expensive parameterization process as well as slow online execution time because of the computationally intensive computation. Besides, the performance of model-based methods declines under uncertainties in battery model parameters as well as noise and aging impacts (Li et al., 2020e). Wei et al. (2016) suggested a novel method for SOC estimation and model identification with noise effect based on Frisch scheme-based bias compensating recursive least squares (FBCRLS). The experimental results demonstrated the effectiveness of the proposed approach in reducing the bias noise on model identification which elevated the SOC estimation accuracy. Wei et al. (2017) designed a dual estimation and multi-time scale based SOC and capacity estimation framework in BMS. The online battery model parameters were assessed using vector-type recursive least squares (VRLS). The dual estimation was carried out through the integration of RLS-based capacity estimator and Kalman filter (KF) based SOC estimation. The experiments report demonstrated the high accuracy and fast convergence in SOC and capacity estimation in real-time. Wei et al. (2019) estimated online SOC and power capacity under the impact of noise corruption. The adaptive forgetting recursive total least squares (AF-RTLS) was employed to lessen the bias and noise effect in model parameters. AF-RTLS was incorporated with Luenberger observer (OB) to online compute SOC. The multi-constraint analytical method together with OB based model parameters and SOC were used to assess the capacity online. The simulation and experiments illustrated the superiority of the proposed approach with regard to accuracy and robustness.

On the other hand, Intelligent algorithms have received huge attention in battery state estimation because they need limited knowledge and less development time to build complex battery systems compared to model-based methods (Y. Song et al., 2020b). Intelligent algorithms and controller schemes do not require comprehensive domain knowledge and detailed mathematical model rather only requires a large pool of data and high computing power (L. Song et al., 2020a). Also, intelligent methods achieve outstanding performance if a sufficient amount of data is available and are very efficient under uncertainties including noise, varying temperatures and aging effects (Singh et al., 2020). Moreover, they have fast online execution and self-learning operations to execute the parameterization (Yao et al., 2021). Nevertheless, they have shortcomings in terms of long training time (Deng et al., 2019).

3.1.1. State of charge estimation

SOC is defined as how much charge is available inside a battery cell to drive a vehicle (Lipu et al., 2020b). The precise evaluation of SOC is very significant in achieving better charging, discharging profile and prolonged battery lifespan (Shu et al., 2020). Numerous articles have been published on SOC estimation for lithium-ion batteries. Based on the literature survey, SOC algorithms generally have four categories: feed-forward algorithms, regression and probabilistic algorithms, time-series algorithms and hybrid intelligent optimization algorithms.

3.1.1.1. Feed-forward neural network algorithm-based SOC estimation. Machine learning-based algorithms can map the battery dynamics without the battery model, as well as complex mathematical rules and relationships. He et al. (2014) proposed the

feedforward neural network (FNN) algorithm with the back-propagation learning rule using three input features, and two hidden layers (HLs) for accurate SOC estimation of $LiFePO_4$ batteries. FNN obtained maximum SOC error of 1.9% and 3.5% under Federal Urban Driving Cycles (FUDS) and US06 cycle, respectively. Although FNN method has simple execution and acceptable accuracy, it needs long data training duration. Kang et al. (2014) constructed an enhanced radial basis function neural network (RBFNN) model with adjusted center, weighted and activation function. RBFNN achieved reasonable results under different aging profiles and EV drive cycles with mean absolute error (MAE) under 5%. RBFNN method can achieve global approximation as well as reasonable accuracy with incomplete information; however, training operation is lengthy and can be trapped in a local minimum. Lipu et al. (2019b) developed an ELM structure with one hidden layer and sigmoid activation function. ELM delivered accurate SOC solutions with low computational cost, reducing MAE by 35.3% and 23.6% compared to RBFNN and FNN methods, respectively under different drive cycles. Although ELM method exhibits better generalization, scalability and fast computation process, the suitable number of HLs and hidden neurons (HNs) is required to achieve better results. Cui et al. (2018) introduced a wavelet neural network (WNN) to monitor SOC for EV batteries. WNN obtained better accuracy and computation time than FNN and EKF algorithms, estimating MAE and maximum SOC error of 0.59%, and 3.13%, respectively. WNN features easy training execution but it needs several hidden layer units to obtain reasonable solutions. Chemali et al. (2018) suggested a deep FNN (DFNN) to compute SOC against Gaussian noise, offset, gains, temperatures, and diverse EV drive cycles. DFNN achieved the highest SOC error and MAE of 8% and 4% respectively, at varying temperature conditions. The configurations of feed-forward neural network algorithms in battery SOC estimation is shown in Fig. 3.

State of energy (SOE) is another assessment index of BMS which is computed in adjacent with SOC. SOE is calculated using the residual energy storage capacity divided by the maximum energy storage capacity in a battery cell (Hu et al., 2019). Liu et al. (2014b) introduced FNN to determine SOE for lithium-ion batteries under the current influence and different temperature settings. FNN achieved acceptable solutions with RMSE of 1.314% against dynamic loading conditions. Dong et al. (2015) proposed WNN-based SOE estimation method for LiFePO4 batteries. WNN demonstrated accurate and robust performance under different discharge current rates and temperature conditions. The particle filter estimator is further applied to eliminate the effects of measurement noises in current and voltage measurements. WNN obtained RMSE of 1.89% and 5.93% under constant current and variant current, respectively.

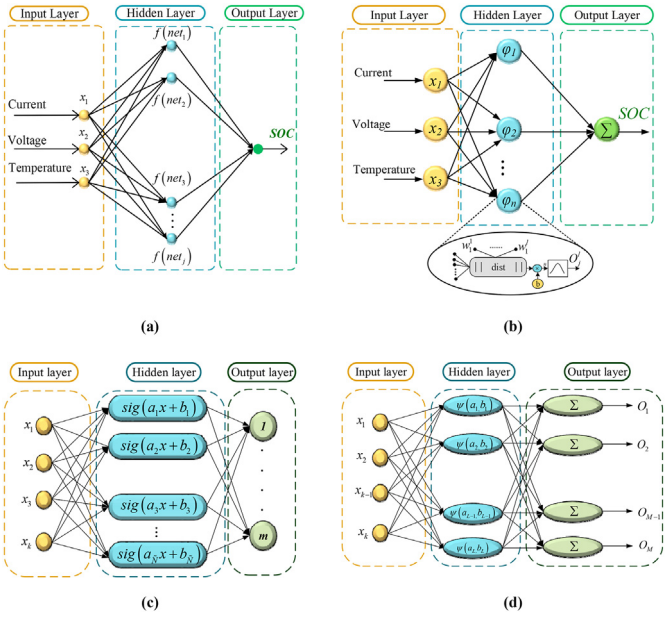


Fig. 3. Structure of feedforward neural network algorithms based SOC estimation (a) FNN (He et al., 2014), (b) RBFNN (Kang et al., 2014), (c) ELM (Lipu et al., 2019b), and (d) GPR (Sahinoglu et al., 2018).

3.1.1.2. Regression and probabilistic algorithm-based SOC estimation. Álvarez Antón et al. (2013) built a support vector machine (SVM) model for SOC estimation of $LiFeMnPO_4$ batteries. SVM delivered a reasonable solution with respect to accuracy and execution time, indicating root mean square error (RMSE) of 0.71%. SVM can address non-linear and high dimension models with less-overfitting issues, however, it has complex computation and time-consuming process. Li et al. (2014a) evaluated random forest (RF) based SOC estimation accuracy under battery discharge currents. RF obtained better results than FNN under steady-state and dynamic state, reporting RMSE of 0.59% and 0.14%, respectively. RF is accurate and robust to over-fitting and noise impacts; nevertheless, its performance deviates under irregular and dynamic systems. Sahinoglu et al. (2018) developed Gaussian process regression (GPR) based SOC estimation approach in electric vehicle batteries. The structure of GPR was enhanced with optimal hyper-parameters and kernel function. GPR outperformed FNN and SVM, attaining MAE and RMSE below 0.81%, and 0.24% respectively in EV drive cycles. Although GPR has easy interpretability of various features and is excellent in capturing the model uncertainty; it suffers from low accuracy in high dimensional spaces.

3.1.1.3. Time series algorithm-based SOC estimation. Time-series based algorithms are computationally rich and very efficient since they predict future SOC estimation based on the present information as well as historical information of input, hidden, and output states. Chaoui et al. (2017) proposed a time-series algorithm for SOC estimation for $LiFePO_4$ batteries based on input time-delay neural network (TDNN). TDNN was validated under four calendar aging cycles and obtained RMSE below 0.33% at 10 A constant discharge test. The recurrent neural network (RNN) algorithms are the basic methods for exploring time-series based algorithms. Chaoui and Ibe-Ekeocha (2017) designed a RNN with the nonlinear autoregressive network with exogenous inputs (NARX) architecture to determine SOC for Lithium Iron Phosphate (LiFP) and Lithium Titanate (LiTO) batteries. The proposed model delivered high accuracy and low computational cost under numerous charge-discharge profiles and aging cycles. The NARX is computationally intelligent in controlling the dynamic and complex system, but the performance depends on the number of historical data points in the input layer, hidden layer, and output layer.

Due to the availability of a huge amount of quality data, large storage capacity, and powerful computation processor, the deeper architectures of RNN have become prevalent. However, deep learning-based SOC estimation is still very limited. Yang et al. (2019b) introduced a stacked long short-term memory (LSTM) network to estimate SOC for $LiFePO_4$ batteries. LSTM illustrated low execution time, and excellent tracking accuracy under erroneous initial SOC values, achieving MAE and RMSE within 1% and 2%, respectively. Bian et al. (2020b) suggested a bidirectional LSTM framework with encoder-decoder to estimate SOC under various temperature settings. The encoder-decoder supported LSTM algorithm in mapping the long-term dependencies in past and future resulting in a rise in SOC estimation accuracy. The experimental

dataset demonstrated the accurate solutions in SOC estimation, indicating MAE of 1.07% at variable temperature conditions. Yang et al. (2020) introduced LSTM network for battery SOC estimation under FUDS and US06 dive cycles under varying temperature conditions. The results indicated MAE and RMSE 1% and 1.1% respectively. Though LSTM exhibits powerful prediction ability and can deliver satisfactory results in dynamic load profiles, it has shortcomings of having slow and complex training operations. Xiao et al. (2019) employed Adam and Ensemble optimizer to build a gated recurrent unit (GRU). The authors found that GRU- Ensemble demonstrated better accuracy and training time than GRU-Adam, estimating MAE and RMSE of 3.28%, 0.84%, and 1.13%, respectively. GRU can tackle the gating mechanism problems of LSTM; but it has weaknesses in terms of a huge amount of data training and large storage device.

In recent times, the deep learning approaches are combined to build a hybrid algorithm based SOC estimation model. For instance, Song et al. (2019) merged LSTM and convolutional neural network (CNN) to estimate SOC for EV batteries. CNN was used to extract features while the LSTM was employed to evaluate SOC performance. CNN-LSTM illustrated better performance than alone LSTM and CNN network achieving RMSE and MAE inside 1.5% and 1%, respectively. The different structures of time-series based SOC estimation are depicted in Fig. 4. The mathematical equations of the execution process for SOC estimation using different intelligent algorithms are presented in Table 1.

3.1.1.4. Hybrid intelligent optimization algorithm-based SOC estimation. Recently, the integration of heuristic optimization techniques and intelligent algorithms has achieved considerable contributions toward SOC accuracy enhancement. A hybrid intelligent optimization algorithm for SOC estimation can be established through the optimization of hyperparameters which eventually improves SOC estimation accuracy under various operating conditions. For instance, Chang (2013) employed orthogonal least-squares (OLS) and genetic algorithm (GA) to optimize HNs and center, width respectively of RBFNN. The enhanced RBFNN algorithm was dominant to FNN with a maximum absolute error of 1.82%. GA has

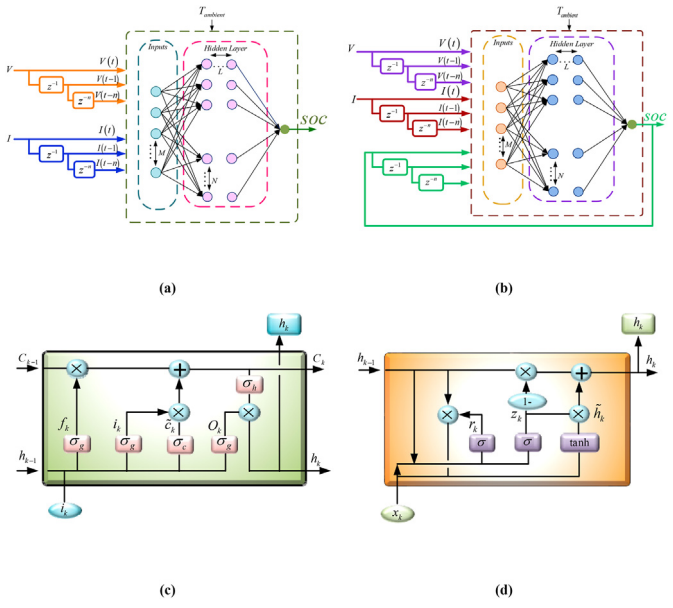


Fig. 4. The structures of time-series algorithm-based SOC estimation (a) TDNN algorithm (Chaoui et al., 2017), (b) Recurrent NARX algorithm (Chaoui and Ibe-Ekeocha, 2017), (c) LSTM layer (Yang et al., 2019b), and (d) GRU layer (Xiao et al., 2019).

an easy execution process but suffers from slow convergence speed. Zhang et al. (2018e) introduced an intelligent method using particle swarm optimization (PSO) to optimize weights and thresholds value of FNN. The optimized FNN improved SOC precision by 2% points in comparison to conventional FNN under dynamic EV drive cycle. Although PSO features good convergence speed and high efficiency, it can converge prematurely. Hannan et al. (2018b) integrated the backtracking search algorithm (BSA) and FNN to determine the suitable number of HNs and learning rate. BSA-based FNN presented better results than conventional FNN, indicating MAE and RMSE of 0.48% and 0.81%, respectively under dynamic stress test (DST) cycle at 25 °C. BSA has a better search exploration process with the mutation and crossover technique; however, it has lengthy computation due to the dual population algorithm. Hannan et al. (2020b) designed an intelligent hybrid optimization model with lighting search algorithm (LSA) based NARX. The accuracy of SOC was improved by optimizing input delays (IDs), feedback delays (FDs), and HNs. An extensive comparative analysis with prominent optimized machine learning algorithms was performed and results were excellent with RMSE and MAE below 1%. LSA offers high exploration and exploitation search capability, nevertheless, it has a complex execution process. Lipu et al. (2019) incorporated the gravitational search algorithm (GSA) and ELM to determine the optimal numbers of the hidden layer (HLs) and HNs. GSA-based ELM was outstanding in delivering SOC in a quick period under US06 cycle, indicating a computational cost of 0.22 s and 0.27 s at 25 °C and 45 °C, respectively. Although GSA features fast learning speed and small training error, it has poor searching capability. Lipu et al. (2020b) introduced an optimized TDNN framework for SOC estimation in lithium-ion batteries based on the firefly algorithm (FA). The improved FA (iFA) was employed to find the optimal hyperparameters including HNs and input time delay. The iFA based RNN delivered excellent solutions in achieving a lower RMSE of 0.8512% and 0.7937% in LiNiCoAlO2 and LiNiMnCoO2 battery, respectively under the Hybrid Pulse Power Characterization (HPPC) test. Although FA is flexible and has an easy implementation, it has shortcomings of slow convergence. The evaluation of intelligent algorithms based SOC estimation is denoted in Table 2.

3.1.2. State of health prediction

SOH is defined as the charge capacity of an aged battery cell to fulfill the operating requirements compared with the charge capacity of a brand-new battery cell (Li et al., 2020b,d). SOH is crucial to assess the current health condition of the battery. Various methods have been published in the literature to predict SOH for lithium-ion batteries. This study divides SOH methods into four groups: feed-forward algorithms, regression and probabilistic algorithms, and recurrent neural network algorithms, entropy algorithms, and other algorithms.

3.1.2.1. Feed-forward neural network algorithm-based SOH prediction. You et al. (2016) developed a real-time FNN based SOH estimation method for EV batteries. FNN was validated by diverse EV drive cycles and attained accurate results, with RMSE and MAE of 1.06% and MAE 3.81% respectively under different temperature settings. Pan et al. (2018) built a novel SOH framework with a multiple correlation analysis-based ELM network. ELM performed better than FNN with MAE and RMSE of 1.09% and 1.72%, respectively. Klass et al. (2014) used on-board available battery data to predict SOH using SVM where SVM achieved RMSE of 0.63% under changing temperature conditions.

3.1.2.2. Regression and probabilistic algorithm-based SOH prediction. Li et al. (2020b) proposed a data-driven algorithm-based battery

Table 1
Mathematical expressions of different intelligent algorithms utilized in SOC estimation.

Refs.	Methods	Algorithms execution formulas	Description
He et al. (2014)	FNN	$f(net) = \frac{1}{1+e^{-(net)}}$ $\begin{cases} net_j = \sum_j w_{i,j}x_i + \theta_{i,j} \\ net_k = \sum_k w_{j,k}O_j + \theta_{j,k} \end{cases}$	net_j, is hidden layer output; net_k is output layer output; x_i is the input vector; wi,j wk,j and θi,j θk,j present the weight and bias, respectively; and O_j represents the output of the hidden layer.
Kang et al. (2014)	RBFNN	$y_k = \sum_{m=1}^{M} w_{km} e^{-\frlVert x(n)-\lambda_m(n)\rVert}{\sigma_m^2(n)}$	λm and σm denote the center and width of m_{th} HNs in the Gaussian function, respectively; wkm denotes weight; yk represents the output in the output layer;
Lipu et al. (2019)	ELM	$\sum_{i=1}^{L} B_i G(a_i, b_i, x_j) = t_j$ $G(a_i, b_i, x_j) = (1 + e^{-(a_i, b_i, x_j)})^{-1}$ $\beta = H^{+}T,\ \beta = [\beta_1, ..., \beta_L]^T, T = [t_1, ..., t_N]^T$	x_j is input weight vector; a_i is hidden weight vector; b_i is hidden layer bias; B_i is output weight; L is HNs; $G(x)$ activation function; tj is the output matrix of the hidden layer; H+ is the Moore −Penrose generalized inverse of H.
Álvarez Antón et al. (2013)	SVM	$y = \alpha^T.\beta(x) + s$	α is the weight; $\beta(x)$ is a mapping to dimensional feature space; s is the expression of biasness.
Li et al. (2014a)	RF	$MSE \approx MSE^{OBB} = \frac{1}{N}\sum_{i=1}^{N}(\hat{Y}(X_i) - Y_i)^2$	MSE^{OBB} is the out-of-bag (OBB) error which represents the mean square error (MSE) of RF; YXi are the predicted output; Yi is the observed output; N is the total number of samples.
Cui et al. (2018)	WNN	$x_l' = \psi_{a_l,b_l}\left(\frac{\sum_{k=1}^{K} w_{kl}x_k - b_l}{a_l}\right)$ $x_m'' = net_m' = \sum_{l=1}^{L} w_{lm}' x_l'$	xl' is the output of the hidden layer; $\psi(x)$ is generating function; a_l and b_l are the wavelet dilation and translator parameters respectively; w_{kl} and wlm' are weight vector in the hidden layer and output layer, respectively; x_k is input layer data; xm'' is the output of the output layer.
Hannan et al. (2020b)	NARX	$y(n+1) = f_0\left[b_0\sum_{h=1}^{Nh} w_{ho} f_h\left(b_h + \sum_{i=0}^{d_u} w_{ih}u(n-d_u) + \sum_{i=0}^{d_y} w_{jh}u(n-d_y)\right)\right]$	wihwih,who who, wjhwjh are the weight vector; b0b0, bh are the bias vector; fh(.)fh(.)fh., $f_0(.)$ are the activation function; d_u and d_y are number of input delay and feedback delay, respectively.
Yang et al. (2019b)	LSTM	$f_k = \sigma_g(W_f x_k + U_f h_{k-1} + b_f)$ $i_k = \sigma_g(W_i x_k + U_i h_{k-1} + b_i)$ $o_k = \sigma_g(W_o x_k + U_0 h_{k-1} + b_0)$ $c_k = f_k \circ c_{k-1} + i_k \sigma_c(W_c x_k + U_C h_{k-1} + b_c)$ $h_k = o_k \circ \sigma_h(c_k)$	x_k is the input of the LSTM unit at time k; $h_k$is the hidden state; $c_k$is the unit memory; W, U, are the weight matrices; b is bias parameter; i_k, $f_k$and $o_k$are the activation function of the input gate, forget gate, and output gate respectively; σ_g, σ_c, and σ_h are sigmoid, hyperbolic tangent, and hyperbolic tangent, respectively.
Xiao et al. (2019)	GRU	$z_k = \sigma(W_z^x x_k + W_z^h h_{k-1} + b_z)$ $r_k = \sigma(W_r^x x_k + W_r^h h_{k-1} + b_r)$ $\tilde{h}_k = \tanh(W_h^x x_k + W_h^h(h_{k-1} \circ r_k) + b_h)$ $h_k = z_k \circ \tilde{h}_k + (1 - z_k) \circ h_{k-1}$	z_k and r_k denote the gate updating and resetting vector, respectively; σstands for sigmoid function; h_k and $\tilde{h}_k$ define the state vector and candidate state, respectively; The parameters during gate updating operation are denoted by W_z^x, W_z^h, and b_k;while the parameters during gate resetting operation are represented by W_r^x, Wrh, and br; the candidate state parameters are depicted by Whx, Whh and b_h.

degradation model to predict SOH using support vector regression (SVR) and ICA. The improved filter method was used to smooth incremental capacity curves followed by the peak fitting technique to decompose the smooth curves. The training dataset was developed using the extracted features from the decomposed incremental capacity curves. The proposed model was validated using the aging datasets and achieved RMSE and MAE under 2%. Li et al. (2014b) designed a reliable prognostic model to examine SOH using a relevance vector machine (RVM). The experimental results indicated an accuracy of 90% with an acceptable limit of error tolerance. Peng et al. (2018) examined battery capacity and SOH based on GPR using the raw data collected from the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland and the Department of Engineering Science (DES) at the University of Oxford. The results illustrated the applicability and stability of GPR with an error of 2.2%. Li et al. (2020c) developed a novel multi-time-scale framework to predict SOH for lithium-ion batteries based on GRP and partial incremental capacity (PIC). First, PIC curves were employed to extract significant features from different degrees such as intercept, peak and slope. Then, kernel-modified GPR was used to build a short-term battery degradation model and accordingly autoregressive long-term battery prediction model was designed. The four-dataset consisting of different health levels and cycling test conditions were applied to verify the effectiveness and robustness of the proposed model and results were promising with regard to battery health situations forecasting. Li et al. (2020d) introduced a novel ICA based GPR model to predict SOH for lithium-ion batteries. GPR method was applied to achieve the smoothing incremental capacity curves. The input features were extracted from the partial incremental capacity curves. The predictive capability of the proposed model was verified using the aging datasets from NASA along with the tested batteries under the different initial health conditions. The validation report illustrated high accuracy and robustness in SOH estimation.

3.1.2.3. Recurrent neural network algorithm-based SOH prediction. Eddahech et al. (2012) designed a RNN network with current, temperature, SOC variation, and constant-phase-element (CPE) to predict SOH by taking into account electrochemical impedance spectroscopy (EIS) measurements. RNN obtained MSE of 0.462% and 0.296% in capacity and resistance prediction. Qu et al. (2019) established a hybrid intelligent method with LSTM and PSO to determine SOH of lithium-ion batteries. PSO based LSTM was validated by the National Aeronautics and Space Administration (NASA) Prognostics Center of Excellence (PCoE) dataset at RT and reported RMSE below 2%. Ungurean et al. (2020) presented a deep learning-based SOH estimation model for lithium-ion batteries using GRU. GRU outperforms LSTM with regard to estimation error and few parameter requirements.

3.1.2.4. Entropy algorithm-based SOH prediction. Hu et al. (2014) proposed sample entropy-based SOH prediction using an aging dataset at three different temperatures. The results were excellent under the HPPC test with an average relative error of 2%. Hu et al.

Table 2
The evaluation results of the intelligent algorithms for SOC estimation in BMS.

Algorithms	Refs.	Input features and output	Structure/functions/ hyperparameters	Verification profiles	Error rate	Research gaps/future works
FNN	He et al. (2014)	Variables = [I (t), I(t-4),…, I(t-4k), V (t), V (t-4),…,V(t-4k), Tem (t), T (t-4),…, T (t-4k), Output = [SOC (t)]	-HLs 2 -HNs 5 -Sigmoid function	−10 °C, 25 °C, 40 °C, and 50 °C temperatures - US06 and FUDS drive cycles	-RMSE ≤ 2.5% (US06) -RMSE ≤ 1.4% (FUDS)	-The complexity of FNN was increased with the inclusion of Kalman filter (KF). -Aging effect was not examined.
RBFNN	Kang et al. (2014)	Variables = [I(t), V (t), Cap (t)] Output = [Voltage (t)]	-Single HL -Gaussian function	−10 °C, 25 °C, and 40 °C temperatures - ECE and UDDS drive cycles under 100, 400, and 700 cycles.	-MAE ≤ 5%	-The network parameters were not optimized. -Computational cost was not mentioned.
ELM	Lipu et al. (2019b)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-Single HL -HNs 147/158/220/328. -Sigmoid function	−25 °C, and 45 °C temperatures -BJDST and US06 drive cycles	-RMSE ≤ 0.76% (BJDST) -RMSE ≤ 1.56% (BJDST)	-The proposed model was tested with only two drive cycle data and two temperatures.
SVM	Álvarez Antón et al. (2013)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-Kernel parameter 0.5 -SVM type regression 0.25	-Different charge and discharge current rates. -Room temperature (RT)	-R^2 0.97	-The model was not validated under different environmental settings and drive cycles.
RF	Li et al. (2014a)	Variables = [I (t), V (t), T (t), DT (t), VC (t), TC (t)] Output = [SOC (t)]	-Number of trees 500 -Number of variables 2	- NASA database - RT	-RMSE 0.14% (dynamic state) -RMSE 0.59% (steady state)	-The outcomes lacked in detailed comparative analysis with other intelligent methods under different operating conditions.
WNN	Cui et al. (2018)	Variables = [I(t), V (t)] Output = [SOC (t)]	-Multi-hidden layer -HNs 10 -Morlet wavelet function	-NEDC, UDDS drive cycle −25 °C temperature.	-MAE < 1% -SOC $Error_{MAX}$ < 3.5%	- Temperature impact was not investigated. -Scaled conjugate gradient and Bayesian regularization can be employed in the future.
GPR	Sahinoglu et al. (2018)	Inputs = [I(t), I (t-1), V(t), V (t-1), T (t), T(t-1), SOC (t-1)] Output = [SOC (t)]	-Single HL -Kernel function	−10 °C, and 25 °C temperatures. -EV load profiles.	-RMSE < 0.3% -MAE < 1%	- The validation was performed without temperature effects. -The influence of battery aging can be considered in the future.
TDNN	Chaoui et al. (2017)	Inputs = [I(t), I (t-1),…, I (t-20), V (t), V (t-1),…,V (t-20), T (t), Output = [SOC (t)]	-HLs 2 -HNs 3/2 (first/second hidden layer) -Sigmoid activation function.	−6 A, 10 A, and 20 A discharge currents. -Ambient temperature	-RMSE ≤ 0.33%	-Validation was limited to only specific charge discharge current rates. -Detailed comparative analysis was not carried out.
LSTM	Chemali et al. (2017)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-HNs 50 -Batch size 250, 500, and 1000 -Sigmoid and tanh	-HWFET, UDDS, LA92 and US06 drive cycles. -RT	-SOC $Error_{MAX}$ ≤ 5.38%	-The hyperparameters of LSTM were chosen randomly. -High configuration graphical processing units (GPU) are required for fast training execution.
GRU	Xiao et al. (2019)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-GRU units 200 -Batch size 230 -Learning rate (LR) 0.001 and 0.0005 -Epochs 100 -Ensemble optimizer -Sigmoid and tanh	−10 °C, 30 °C, and 50 °C temperatures. -FUDS, DST and US06 drive cycles.	-RMSE 2.45% (FUDS) -RMSE ≤ 1.97% (DST) -RMSE 1.40% (US06)	-The hyperparameter selection of the GRU was based on experience. -Robustness validation under unknown initial SOC was not performed. - High processing power and memory footprint are required.
GA	Chang, (2013)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	- RBFNN - HNs 24 -The center, and width of the activation function are optimized.	- 0.3C, 0.5C and 1 C discharge currents - RT	-Maximum absolute error of 1.82%.	-The model became complex with three algorithms: RBFNN, GA and orthogonal least square. -EV load profiles and varying temperatures were not investigated.
BSA	Hannan et al. (2018b)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-FNN configuration -The HNs and LR are optimized	−0 °C, 25 °C, and 45 °C temperatures. -FUDS and DST drive cycles.	-RMSE ≤1.74% (FUDS) -RMSE ≤ 1.47% (DST)	- The training process was slow due to the selection of a dual population of BSA. -Comprehensive experiments are required on different types of lithium-ion batteries.
GSA	Lipu et al. (2019b)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-ELM structure -The HNs are optimized.	−25 °C, and 45 °C temperatures. -BJDST and US06 drive cycles.	-RMSE ≤ 0.76% (BJDST) -RMSE ≤ 1.56% (US06)	- GSA can be trapped in local minima. - The proposed model requires real-time evaluation through a prototype embedded system.
LSA	Hannan et al. (2020b)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-NARX model -The IDs, FDs and HNs are optimized	−0 °C, 25 °C, and 45 °C temperatures. - FUDS and US06 drive cycles.	-MAE ≤ 0.58% (FUDS) -MAE ≤ 0.38% (DST)	-Implementation of LSA into NARX was not easy due to several parameters and three projectiles. - The proposed model needs a prototype hardware verification using modularized design.
FA	Lipu et al. (2020b)	Variables = [I(t), V (t), T (t)] Output = [SOC (t)]	-TDNN configuration -The HNs and input time delay are optimized.	-HPPC load profile.	-RMSE 0.8512% ($LiNiCoAlO_2$) -RMSE 0.7937% ($LiNiMnCoO_2$)	- FA suffers from slow convergence speed. - The model robustness needs to be checked under temperature and aging impacts.

I: Current, V: Voltage, T: Temperature, Cap: Capacity, DT: Discharge time, VC: Voltage change, TC: Temperature change, NEDC: New European Driving Cycle; HWFET: Highway Fuel Economy Test, UDDS: Urban Dynamometer Driving Schedule, BJDST: Beijing Dynamic Stress Test, SDT: Static Discharge Test.

(2016a) designed a SOH estimation approach using sparse Bayesian predictive modeling (SBPM) based sample entropy algorithm. The verification and comparison of the proposed model were carried out through an experimental platform using lithium-ion batteries at three different temperatures. The results indicated an improvement of 15.6%, 6.7%, and 0.7% in achieving average relative capacity errors compared to the polynomial model at 10 °C, 22 °C, and 35 °C, respectively. The entropy method can assess the data fluctuation and predictability but has drawbacks with regard to high computational complexity.

3.1.2.5. Other algorithms. Yang et al. (2017) presented GA based SOH prediction model to identify capacity and initial SOC accurately. The experimental cases proved the stability, accuracy, and robustness of the proposed method under different aging states, and results indicated maximum capacity error and SOC error below 5% and 2.1%, respectively. Zhu et al. (2016) considered the battery internal resistance to monitor SOH based on Grey-Markov Chain (GMC) algorithm. GMC was validated through life-cycle experimental tests using 16 battery cells with 3,200 groups of data. The results demonstrated the effectiveness of GMC in predicting capacity and SOH for $LiFePO_4$ battery under different state levels. Comparative performance analysis of intelligent algorithms-based SOH estimation is depicted in Table 3.

3.1.3. Remaining useful life prediction

RUL is the remaining number of load cycles or time of a battery cell to meet its performance requirement (Zhou et al., 2020). RUL is a critical parameter that supports online decision-making activities such as fault identification and diagnosis (Li et al., 2020f). The accurate estimation of RUL is rarely feasible due to the numerous uncertainties which affect such future prediction (Zhang et al., 2019b). Hence, the researchers have been trying to develop an efficient RUL estimation method considering the uncertainties related to prediction and prognostics (Zhou et al., 2020). In this study, RUL prediction methods are divided into machine learning-based algorithms, time-series based algorithms, statistical data-driven algorithms, Monte Carlo algorithms, and other algorithms.

3.1.3.1. Regression algorithm-based RUL prediction. Wang et al. (2014a) designed RVM based SOH prediction model utilizing the capacity degradation model of lithium-ion batteries. RVM was examined by randomly selected aging cycles using three lithium-ion battery cells at ambient temperature. RVM provided prognostic solutions with reasonable absolute percentage error. Wang et al. (2013) employed PSO to find the optimal parameters of the autoregressive (AR) model toward accurate RUL prediction of lithium-ion batteries. The absolute prediction errors of 9 cycles, 7 cycles, 5 cycles, 0 cycles, 8 cycles and 1 cycle were noted in 104, 128, 146, 157, 170, and 183 inspection cycles. Xue et al. (2020) proposed an improved SVR model optimized by GA toward accurate prediction of RUL of lithium-ion batteries. GA was used to find the optimal key parameters of SVR. The GA optimized SVR was tested by NASA battery dataset and achieved better RUL prediction outcomes in comparison to other methods.

3.1.3.2. Recurrent neural network algorithm-based RUL prediction. Zhang et al. (2018d) proposed LSTM to predict RUL under different current rates and temperature settings. LSTM was examined by four lithium-ion cells under different current rates (1 C, 2 C, and 3.5 C) at 25 °C and 40 °C temperatures. The validation reports indicated the superiority of LSTM over SVM and simple RNN. Liu et al. (2019) introduced a data-driven ensemble prognostic approach based on combined Bayesian model averaging (BMA) and LSTM. The CALCE dataset was used for validation and the proposed model demonstrated effectiveness in terms of uncertainty management and prediction performance. Li et al. (2019b) combined the LSTM with Elman neural network (ENN) and EMD to develop a novel hybrid approach to predict RUL. The EMD was applied to decompose the battery capacity curve into some sub-layers while the LSTM and ENN were employed to forecast the high and low-frequency sub-layers, respectively. A comprehensive experimental test was conducted to evaluate the performance and results were promising, indicating relative error under 3.5%. Ma et al. (2019) proposed false nearest neighbors method-based hybrid neural network method to predict RUL for lithium-ion batteries. The hybrid neural network was structured using CNN and LSTM algorithms. The false nearest neighbors approach was employed to determine the optimal window size while hybrid neural network was used to conduct the model training and testing. The comparative analysis with other state-of-the-art methods proved the effectiveness of the proposed method in forecasting RUL with lower error rate.

3.1.3.3. Statistical data-driven algorithm-based RUL prediction. Li et al. (2016) proposed a Gaussian process mixture (GPM) based novel RUL prediction technique of lithium-ion battery. The results illustrated that GPM was more accurate and robust than SVM with regard to RMSE, indicating 1.58% and 1.3% at 60th and 80th cycles, respectively. Tang et al. (2014) proposed RUL prognostic strategy based on the Wiener process with measurement error (WPME). The practical case studies, as well as numerous examples, illustrated the superiority of WPME over the traditional maximum likelihood estimation (MLE) in terms of accuracy and robustness.

3.1.3.4. Monte Carlo algorithm-based RUL prediction. Xiong et al. (2019) employed a moving-window-based method to indicate lithium-ion battery RUL using Monte Carlo (MC) simulation. The experiments were conducted using 1 C and 2 C current rates under 25 °C and 40 °C temperatures and results reported capacity estimation errors within 1.5%. Zhang et al. (2019a) employed MC simulation to develop an enhanced RUL prognostic model for lithium-ion batteries using the Box-Cox transformation (BCT). An accurate RUL prediction was obtained with standard deviations and error of [1.8, 7] cycles, and [−20, 10] cycles, respectively. MC can map the non-linear, complex problems and provide satisfactory solutions in probability distributions but it needs high computation powerful processor.

3.1.3.5. Other algorithms. Ng et al. (2014) established a naive bayes (NB) based degradation model of lithium-ion battery to evaluate RUL at different temperatures and discharge current rates. NB delivered better predictions than SVM with an average RMSE of 16.1 cycles (0.17%), and SD of 10.7 cycles (0.15%). NB has an easy execution process and short computation time; nonetheless, it has dependencies among variables. He et al. (2011) introduced the Dempster−Shafer Theory (DST) and Bayesian Monte Carlo (BMC) techniques to examine RUL using the double-index degradation model. DST based BMC predicted RUL accurately with SD of 12 and 63 cycles in case one and case two respectively. The performance analysis of intelligent methods for RUL prediction is presented in Table 4. The performance analysis of state estimation in BMS under different case studies including temperature, noise, unknown initial SOC and aging cycles is depicted in Table 5. The advantages and disadvantages of intelligent algorithms in battery state estimation are shown in Table 6.

3.2. Battery equalization in BMS

The battery pack in EV is typically configured using multiple

Table 3
The assessment results of intelligent approaches for SOH prediction in BMS.

Algorithms	Refs.	Input features and output	Structure/ functions	Verification profiles	accuracy	Research Gaps/Future Works
FNN	You et al. (2016)	Variables = [I (t), V (t), T (t)] Output = [SOH (t)]	-HLs 2 -HNs 80	−10 EV drive cycles −10 °C, 25 °C, 45 °C, and 60 °C temperatures.	-RMSE 1.06% -MAE 3.81%	-Rest impact on the aging process need to be considered. -The proposed method can be extended to evaluate SOH for the lithium-ion battery packs.
ELM	Pan et al. (2018)	Variables = [I (t), V(t), T (t), (R (t), P_R (t)] Output = [SOH (t)]	-Single hidden layer -HNs 20 -Sigmoid function	-NEDC, UDDS, and Japan Drive Cycle (JP1015) - 25 °C ± 2 °C temperature.	-RMSE 1.09% -MAE 1.72%	-The effect of low temperature on the resistance can be considered further. -SOH prediction for battery pack comprising series-parallel combination needs to be studied.
SVM	Klass et al. (2014)	Variables = [I(t), V (t), T (t), SOC (t)] Output = [SOH (t)]	-SVM Kernel option 14 -Radial based kernel function	-EV current profiles −0°, 10°, 20°, 30° and 40 °C temperatures.	-RMSE 0.85% (temperature-independent) -RMSE 0.63% (temperature -dependent)	-SVM faces difficulties in calculating the loss function and penalty factor. -The diffusion resistance factor for drive cycles need to be explored.
RVM	Li et al. (2014b)	Variables = [I (t), V (t), Cap (t)] Output = [SOH (t)]	-Polynomial kernel function	−500,600,700, and 800 aging cycles -Ambient temperature (25 °C)	-RMSE $\leq$ 0.357% -$R^2 \leq$ 0.98	-Complexity of the nonlinear mapping was raised with the forecast horizon. -The battery failure and preventive maintenance need to be considered to enhance the reliability of the battery system.
GPR	Peng et al. (2018)	Variables = [I (t), V (t), Cap (t)] Output = [SOH (t)]	-Single hidden layer -Kernel function	-CALCE dataset -DES −40 °C temperature.	-SOC error <7%	-The proposed model needs to be updated under dynamic conditions. -The outcome of this model can be compared with the RVM and model-based approaches.
RNN	Eddahech et al. (2012)	Variables = [I (t), Temperature (t), SOC (t), R/C (t), R/C (t-1), R/C (t-2)] Output = [SOH (t)]	- HNs 6. - Prediction window 3 -Sigmoid activation function.	-ECE 15 urban driving cycle - 40 °C, 45 °C, and 50 °C temperatures.	-MSE 0.462% (capacity) -MSE 0.296% (resistance)	-The real-time automated system for observing battery performance and predicting SOH needs to be developed.
LSTM	Qu et al. (2019)	Variables = [I (t), V (t), Cap (t)] -Output = [SOH (t)]	-HNs 64 -Batch size 20, 29, and 35 -Window size 9, 10, and 11 -Sigmoid and tanh	-NASA PCoE -RT	-RMSE $\leq$ 2%	-The proposed model needs to be assessed for a higher number of cycles. -The adaptability of the method needs to be verified using multiple cells and dynamic load conditions.
GRU	Ungurean et al. (2020)	Variables = [Cap (t)] -Output = [SOH (t)]	-GRU layers 2 -GRU unit 50 -Window size 5 -Epoch 150 -Dropout 0.5 -Sigmoid and tanh	-NASA PCoE −4 °C, 25 °C, and 44 °C temperatures.	-SOC $Error_{MAX}$ < ±6%	-Key parameters including, hidden units, batch size, weight and bias are not optimized. -The embedded implementations need to be carried out.
Entropy	Hu et al. (2014)	Variables = [I (t), Voltage (t), T (t), Cap (t)] Output = [SOH (t)]	-Window length 1, and 2 -Tolerance between 0.1 and 0.25 -Length 100 to 5,000 data points	-HPPC load profile −10 °C, 22 °C and 35 °C temperatures.	-SOC $Error_{MAX} \leq$ 2%	-Validation can be compared other optimization algorithm for better result.

R: Ohmic resistance, C: Capacitance, P_R: Polarization resistance.

batteries connected either in series or parallel to fulfill the power and energy supplies of EVs. Nevertheless, due to the aging influence, ambient temperature variation, and accordingly change in battery internal resistance and capacitance result in battery inconsistency issues which reduce the safety, lifespan, effectiveness of battery pack (Shen and Gao, 2019) (Huang and Abu Qahouq, 2015). Hence, the development of an efficient and reasonable equalization control strategy is an urgent necessity to address these battery inconsistency problems. The equalization control method of the battery pack is divided into groups; dissipation equalization and non-dissipation equalization approach (Y. Chen et al., 2016; Shang et al., 2017a) The dissipation equalization approach is structured using parallel resistors in every single battery and exhibits shortcomings in terms of low efficiency, large power consumption, heat dissipation, and slow equalization speed (Baronti et al., 2014). In contrast, the non-dissipation equalization approach is designed using energy transfer circuits and features high efficiency, low energy consumption, and fast equalization (Shang et al., 2017b). Generally, the battery voltage, capacity, and SOC are employed as the equalization variables, as shown in Fig. 5 (Feng et al., 2019). Battery voltage-based equalization has easy implementation but it is affected by the internal parameters (Shang et al., 2018). Battery capacity-based equalization has the benefits of utilizing the maximum capacity, however, it cannot be appropriate for online operation (Diao et al., 2018). Battery SOC-based equalization has positives with regard to battery pack safety improvement, and energy optimization but requires complex algorithms and a powerful controller to achieve precise solutions (Li et al.,

Table 4
The comparative study of intelligent methods for RUL prediction in BMS.

Algorithms	Refs.	Input features and output	Structure/functions	Validation Profiles	Accuracy	Research Gaps/Future Works
RVM	Wang et al. (2013)	Inputs = [Cap (t)] Output = [RUL (t)]	-Polynomial kernel function	-Randomly selected inspection cycles. - Ambient temperature (25 °C)	-Absolute prediction error 4,3,7,2 cycles at 10, 117, 176, 190, cycles respectively	-Further research can be conducted under a higher number of aging cycles.
LSTM	Zhang et al. (2018d)	Inputs = [Cap (t)] Output = [RUL (t)]	-LSTM layer 2 -HNs 50, and 100 -Retained probability 0.2 -Sigmoid and tanh	−1 C, 2 C and 3.5 C discharge current under maximum 600 aging cycles −25 °C, and 40 °C temperatures.	-SD 28 (253 cycle for cell 1) -SD 24 (285 cycle for cell 2)	-The proposed model needs to be evaluated under EV drive cycle data.
NB	Ng et al. (2014)	Inputs = [Cap (t)] Output = [RUL (t)]	-Discriminant function -Regression Probability distribution −10-fold cross-validation (CV) in training	-NASA test data -1C and 2C discharge current -RT (25 °C) temperature.	-Average RMSE of 16.1 cycles (0.17%) -SD of 10.7 cycles (0.15%).	-The accuracy of the proposed model needs to be checked under dynamic tests and varying temperature settings.
GPM	Li et al. (2016)	Inputs = [Cap (t)] Output = [RUL (t)]	-Gaussian covariance function -Kernel function -Maximum likelihood algorithm -Fixed mixing proportions as the gating function	-NASA PCoE -RT (25 °C) temperature.	-RMSE 1.58% at 60th cycles -RMSE 1.3% at 80th cycles	-Future research work can be conducted on improving training speed.
WPME	Tang et al. (2014)	Inputs = [Cap (t)] Output = [RUL (t)]	-Inverse Gaussian distributions	-NASA test data -RT (25 °C) temperature.	MSE <0.1%	-Nonlinear degradation trend and relaxation effect can be take into consideration for future research.
MC	Zhang et al. (2019a)	Inputs = [Cap (t)] Output = [RUL (t)]	-Transformation parameter −6.53 and −8.72 for Cell C and Cell D respectively. -Pearson correlation coefficient −0.9888	−0.5C, 1C, and 2C discharge current −10 °C, 25 °C, and 40 °C temperatures.	-SD 4.2 cycles (Cell A) -SD 7.2 cycles (Cell C) -SD 3.9 cycles (Cell E)	-The acceleration of aging time using three or more battery cells under test conditions need to be monitored.

2019a). This review categorizes the battery equalization methods into two parts; rule-based algorithms and optimization-based algorithms.

3.2.1. Rule-based algorithms

Usually, various kinds of converter circuits such as buck-boost, Ćuk converters, fly-back converter along with controller; fuzzy logic, proportional integral derivative (PID) are deployed for battery equalization to achieve better scalability, shorten equalization time and high efficiency. Wang et al. (2020a) designed a buck-boost equalization circuit to minimize the cell inconsistency of series-connected lithium-ion batteries using voltage and SOC as equalization variables. FLC was employed to minimize equalization time and enhance efficiency. The proposed method improved the equalization time and efficiency by 49% and 4.88% under static, charging, and discharging situations. Ma et al. (2018) proposed FLC based equalization scheme to address the inconsistency in series-connected battery cells using a two-stage bidirectional equalization circuit. Simulation results demonstrated the superiority of FLC over the mean-difference method, reducing SOC deviation and equalization duration by 18.5% and 23%, respectively under NEDC drive cycle. Nguyen et al. (2014) focused on designing individual cell equalizers (ICEs) based on the adaptive neuro-fuzzy algorithm. The proposed model is executed in a 0.13- μm CMOS technology and obtained a reasonable learning accuracy error of $1.8 \times e^{-5}$. Liu et al. (2017a) presented model predictive control (MPC) to minimize the SOC imbalance and energy dissipation of the lithium-ion battery pack. The MPC terminated the balancing when SOC difference and energy dissipation reached 4%, and 7% respectively, indicating 93% efficiency. Zhang et al. (2015) introduced a novel intelligent control scheme for battery equalization based on FL-NN algorithm. The authors used a two-stage DC/DC converter topology and FL control to achieve the static SOC. Then, NN was used to obtain the dynamic SOC on the basis of static SOC. The proposed method illustrated enhanced performance with regard to energy efficiency and time efficiency.

3.2.2. Optimization-based algorithms

Zhang et al. (2015) established an intelligent battery equalization control method with fuzzy logic control−neural network (FL-NN) integrated with GA optimization. The results reported that the GA based equalization scheme improved the equalization time and efficiency by 1002.5 s and 93.1%, respectively, compared to the mean-difference algorithm. Sun et al. (2015) designed a novel SOC based equalization algorithm for lithium-ion batteries using PSO. The experimental results denoted that SOC based equalizer achieved maximum SOC gap within 2% and improved the capacity by 13.2%. Liu et al. (2017b) proposed an equalization strategy for the series-connected battery cells based on ant colony optimization (ACO). ACO achieved satisfactory solutions in terms of minimizing energy loss as well as achieving high balancing efficiency and high speed. Table 7 denotes the detailed comparative study of the equalization methods concerning variables, objectives, strengths, and weaknesses.

3.3. Battery fault diagnosis in BMS

Battery safety is of great importance to guarantee the secure, reliable, and safe operation of EVs. Thus, the development of a suitable diagnostics and fault handling mechanism is very significant since a minor fault could ultimately cause severe problems to battery health (Tran and Fowler, 2020). Based on the literature

Table 5
Evaluation of case studies of state estimation in BMS.

Cases	Refs.	Settings	Battery state estimation (SOC/SOH/RUL)	Intelligent algorithm	Battery type	Dataset	Results
Temperature	Chemali et al. (2018)	−20 °C, −10 °C, 0 °C, 10 °C, and 25 °C	SOC	DFNN	−2.9 Ah Panasonic 18650 PF lithium-ion battery	-US06 -HWFET	-MAE ≤ 3.4% (HWFET) -MAE ≤ 2.2% (US06)
	Bian et al. (2020)	0 °C, 10 °C, and 25 °C	SOC	LSTM	−2.9 Ah Panasonic 18650 PF lithium-ion battery	-HWFET	-SOC $error_{MAX}$ ≤ 4.31% -MAE ≤ 1.09%
	Ungurean et al. (2020)	−4 °C, 25 °C, and 44 °C	SOH	GRU	−1.85 Ah 18650 lithium-ion battery	-NASA PCoE	-SOC $error_{MAX}$ < ±6%
	Zhang et al. (2019a)	−10 °C, 25 °C, and 40 °C	RUL	MC	−2.7 Ah $LiNiCoAlO_2$ lithium-ion battery	-Capacity degradation dataset	-MAE 20.7 cycle (Cell A) -MAE 53.7 cycle (Cell C) -MAE 10.6 cycle (Cell E)
Noise	Cui et al. (2018)	-The random noises are added -Positive and negative bias noises are included	SOC	WNN	−2.15 Ah Samsung ICR-18650−22P lithium-ion battery	-UDDS -NEDC -UKBC	-MAE 0.92% (random noise) -MAE 1.16% (bias noise)
	You et al. (2017)	-Inclusion of noise in current and voltage with the aging cycles	SOH	LSTM	−3.1 Ah 18650 lithium-ion battery	- UDDS - LA92DDS	-Average SOC error 2.46%
Unknown initial SOC	Song et al. (2019)	- 100%, 80%, 60%, 40% and 20% initial SOC	SOC	CNN	−1.1 Ah 18650 lithium-ion battery	-DST -FUDS -US06	-MAE ≤ 0.87% -RMSE ≤ 1.38%
	Sahinoglu et al. (2018)	−50% initial SOC	SOC	GPR	−4.93 Ah $LiMn_2O_4$ lithium-ion battery	-Constant charge/discharge test	-MAE 0.54% -RMSE 0.27%
Aging	Hannan et al. (2020)	−50, 100, 150, and 200 aging cycles	SOC	NARX	- 2.0 Ah Samsung ICR18650-26F lithium-ion battery	-HPPC	-MAE ≤ 0.96% -RMSE ≤ 1.28%
	Chaoui et al. (2017)	−0 h, 352 h, 544 h and 650 h calendar aging profiles	SOC	TDNN	- 20 Ah $LiFePO_4$ lithium-ion battery	-Charge and discharge current profiles	-RMSE ≤ 0.33%
	Li et al. (2020d)	−50, 100 and 150 aging cycles	SOH	GPR	−1.85 Ah 18650 lithium-ion battery	-NASA PCoE	-MAE ≤ 1.2% -RMSE ≤ 1.8%
	Li et al. (2020b)	−1,30,60,90,120 and 150 aging cycles ($LiNiMnCO_2$) −1,80,160, 240, 320 and 400 aging cycles ($LiFePO_4$)	SOH	SVR	- $LiNiMnCO_2$ lithium-ion battery - $LiFePO_4$ lithium-ion battery	-Aging cycle test	-MAE ≤ 1.003% ($LiNiMnCO_2$-Model 1) -MAE ≤ 1.642% ($LiFePO_4$-Model 1)
	Li et al. (2019b)	−100, 200, 300, 400, 500, 600 and 700 aging cycles	RUL	LSTM	−1.1 Ah $LiCoO_2$ lithium-ion battery	-CALCE dataset	-RMSE 0.915 (0.5C) -RMSE 0.974 (1C)
	Li et al. (2020c)	−50, 100 and150 aging cycles	RUL	GPR	−1.85 Ah 18650 lithium-ion battery	-NASA PCoE	-MAE 8.51 cycle -RMSE 11.62 cycle

review (Chao et al., 2017; Hendricks et al., 2015; Lyu et al., 2019), The fault mechanisms in BMS are usually very complicated and can be classified into three groups; sensor fault, actuator fault, and battery fault, as depicted in Fig. 6. Sensor faults in BMS cover the faults in the current, voltage, and temperature sensors (Lombardi et al., 2014). The current sensors faults can deviate the estimation accuracy of SOC, SOE, SOH and RUL. Moreover, the battery must be functioned within the safe limit of voltage and temperature recommended by the manufacturer (Lu et al., 2013). If the measured values exceed the boundary, the performance of the battery may deteriorate or even lead to accidents. Besides, sensor faults in voltage and temperate could result in TMS error or battery equalization error in BMS.

Actuators faults in BMS include fuse fault, high voltage contactor fault, controller area network (CAN), bus fault, terminal connector fault, cooling system fault (Liu et al., 2015). Battery connection fault may result in a rise in resistance, produce extreme irregular heat, which further accelerates temperature (Feng et al., 2014; Han et al., 2014; Liu et al., 2014a). Besides, the poor connection between batteries may deliver insufficient power supply which leads to a melting of the battery terminals and risk of accidents (Yao et al., 2015). If the cooling system does not work satisfactorily, the battery cannot operate within the temperature range, hence resulting in thermal runaway.

The battery faults are considered as the critical faults in BMS which take place due to overheating, overcharging, over-discharging, internal short circuit (ISC), external short circuit (ESC), battery swelling, electrolyte leakage, and thermal runaway. The overcharging and over-discharging could cause side reactions, health degradation which eventually result in battery swelling and electrolyte leakage. Battery overcharge and over-discharge could lead to various undesirable battery side reactions, resulting in accelerated degradation. These side reactions and gases generated by the chain reactions during the thermal runway may eventually introduce battery swelling and electrolyte leakage (Bandhauer et al., 2011). The ISC and ESC are occurred due to mechanical collisions, and separator failure caused by manufacturing defects (Yang et al., 2019c). Battery overheating is the direct cause to trigger thermal runway which damages the battery with regard to the melting of the separator, and decomposition of cathode, solid electrolyte interface (SEI), and electrolyte (Wang et al., 2012). Lamb et al. (2015) explored the failure propagation and thermal runway of the multi-cell lithium-ion battery pack. Feng et al. (2016) adopted four preventive measures of thermal runway such as reduction of accumulated energy, improvement of heat dissipation, the inclusion of thermal resistance between batteries, and the rise of onset temperature. Hofmann et al. (2017) revealed that the reduction of battery pressure could eventually delay thermal

Table 6
The benefits and drawbacks of different intelligent algorithms in battery state estimation of BMS.

Methods	Advantages	Disadvantages	Refs.
FNN	- This method has an easy execution. - It has acceptable accuracy.	- It has a lengthy data training process.	He et al. (2014); Liu et al. (2014b); You et al. (2016)
RBFNN	- This method can achieve global approximation. - It obtains reasonable accuracy with incomplete information.	- The training operation has local minimum trapping issues. - It requires a long training duration.	Haddad Zarif et al. (2014); Chen et al. (2016a)
ELM	- This method has a quick computation process. - It has better generalization and scalability.	- The performance of ELM relies on the optimal number of HNs.	Pan et al. (2018)
SVM	- This method has less-overfitting issues. - It has excellent outcomes in non-linear and multi-dimension models.	- It has complex computation - It has a time-consuming operation.	Li et al. (2014b); Álvarez Antón et al. (2013)
RF	- This method has quick and accurate execution under huge datasets. - It is accurate and robust to over-fitting and noise impacts.	- Unable to provide satisfactory solutions in a highly irregular system. - The values of trees and leaves influence the estimation and prediction outcomes.	Li et al. (2014a); Chen et al. (2018)
WNN	- This method has less complexity in selecting hidden nodes and weights. - It has easy training execution.	- Several hidden layer units are needed to obtain reasonable solutions.	Cui et al. (2018); Dong et al. (2015)
GPR	- This method is excellent in capturing the model uncertainty. - It has simple interpretation of various features.	- Poor accuracy in multi-dimensional spaces.	Sahinoglu et al. (2018); Peng et al. (2018)
NARX	- This method is very effective to address time series problems. - It has intelligent computation in controlling the dynamic system.	- The effectiveness relies on the appropriate selection of IDs, FDs and HNs.	Chaoui et al. (2017); Eddahech et al. (2012)
LSTM	- This method has satisfactory outcomes under long-term dependencies. - It needs a lower computational effort during the online stage.	- It needs long and complex training execution. - It requires a costly processor to accelerate training operations.	Chemali et al. (2017); Qu et al. (2019); Zhang et al. (2018d)
GRU	- This method overcomes the gating mechanism issues of LSTM. - It can map the long-term sequential dependencies.	- It wants a huge amount of training data. - It needs a high-performance processor.	Xiao et a. (2019); Ungurean et al. (2020)
Entropy	- This method can quantify the predictability, complexity, and fluctuation of the measured signals.	- It has a high computational complexity.	Hu et al. (2014)
NB	- This method has a short computation time. - It has an easy implementation process.	- It has dependencies among variables. - It needs a large number of records to achieve acceptable solutions.	Ng et al. (2014)
WPME	- This method can handle degradation function and noise.	- It has a high mathematical computation process. - It has slow convergence.	Tang et al. (2014)
MC	- This method can model complex problems. - It has an excellent solution in probability distributions. - It is intuitive and relatively easy to execute.	- This method is costly. - It needs high computation power.	Zhang et al. (2019a)
BSA	- This algorithm has enhanced the exploration capability process with the mutation and crossover technique.	- It has a lengthy computation due to the dual population algorithm. - The search direction and crossover process are complex.	Hannan et al. (2018b)
PSO	- This method has a simple execution. - It needs fewer parameters to adjust. - It has high efficiency and fast convergence speed.	- This method can converge prematurely. - Difficult to outline the initial design parameters.	Zhang et al. (2018e); Xia et al. (2018a)
GSA	- This algorithm has easy execution and fast learning speed. - It has a small training error and better generalization performance.	- This algorithm has a poor searching capability.	Lipu et al. (2019)
LSA	- This algorithm has high exploration and exploitation search capability. - It has high efficiency and good convergence characteristics.	- It has a complex computation process.	Hannan et al. (2020b)
FA	- This algorithm is flexible and has easy execution. - It can provide correct results using a fewer number of iterations.	- It has a limitation of slow convergence.	Lipu et al. (2020b)

runaway propagation.

The methodological framework of fault diagnosis in BMS is illustrated in Fig. 7 which includes data acquisition, feature extraction, fault diagnosis, fault prognosis, and fault handling. First, the data acquisition is employed to collect and store battery data including current, voltage, and temperature from experimental arrangements or test bench models. After, the feature extraction is carried out for data preprocessing, extraction, and selection. Then, the selected data is processed for fault diagnosis and fault prognosis phases. Fault detection, isolation, and estimation are covered under the fault diagnosis process. The fault prognosis stage includes battery state algorithm execution and validation to deliver early detection or prediction of battery faults. Finally, the fault handling stage examines and assesses the outcomes from fault diagnosis and fault prognosis, and accordingly delivers effective decisions including alarm generation, faulty batteries isolation, and power supply shut down. This study divides the fault diagnosis approaches into five sections; rule-based algorithms, machine learning-based algorithms, mathematical data-driven algorithms, entropy algorithm and other algorithms.

3.3.1. Rule-based algorithms

Xiong et al. (2012) suggested a probabilistic and a rule-based technique to detect the failure in lithium-ion batteries. The Boolean expression was formulated based on rules established while the voltage was dropped and the temperature was increased during the battery over-discharge stage. The proposed method was verified by experiments and executed in real-time to identify the faults in over-charged batteries. Muddappa and Anwar (2014) proposed a fault diagnosis method using fuzzy logic for high power battery packs in EV applications. First, residuals with respect to SOC, voltage, and temperature were generated using a model-based observer. Then SOC, voltage, temperature change together with the residuals were integrated into the fuzzy logic algorithm to identify the various faults such as battery aging, overcharge, over-discharge, thus providing a reliable, fast, and effective strategy in

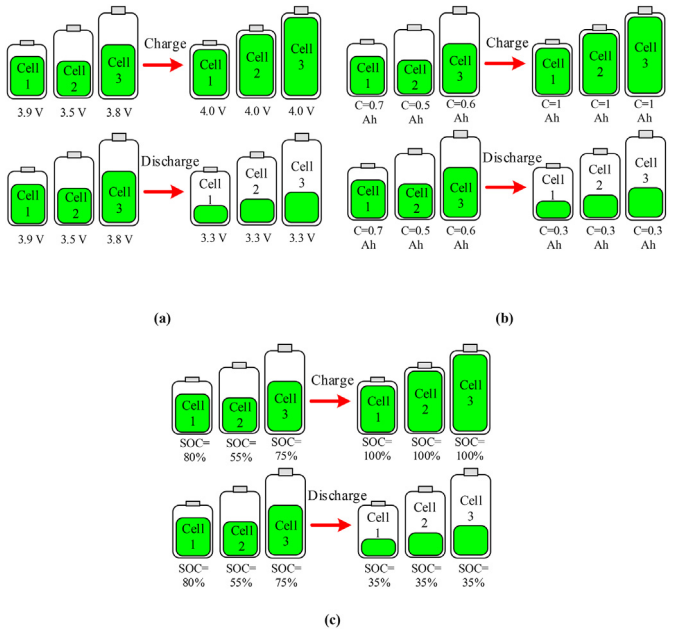


Fig. 5. Battery pack equalization during the charging/discharging control based on (a) voltage, (b) capacity and (c) SOC (Feng et al., 2019).

diagnosing battery faults.

3.3.2. Machine learning-based algorithms

Yang et al. (2018) presented RF-based classification method and three steps model-based diagnosis algorithm to diagnose and prognosis the electrolyte leakage and ESC fault. The pre-trained RF classifier used the RMSE indicator to confirm the leakage condition. The several cases demonstrated the effectiveness of the proposed method in diagnosing ECS battery cells. Zhao et al. (2017) focused on developing a novel fault diagnosis approach for EV batteries using 3σ multi-level screening strategy (3σ-MSS) and machine learning algorithm. The 3σ-MSS was utilized to build the criteria for the identification of the abnormal changes of battery terminal voltages while the machine learning algorithm was employed to execute the diagnosis process using the big data statistical regulation. Hong et al. (2019) developed a novel LSTM algorithm for battery fault diagnosis by taking into account the battery performance concerning weather and driver's behavior in EV operation. All hyperparameters of LSTM were pre-optimized offline using a dual-model-cooperation prediction technique. The prognosis stability and feasibility of the method were examined using real-time data under various voltage abnormities. The results showcased the successful execution of the proposed method in battery protection and safety toward the identification of advent battery faults.

3.3.3. Mathematical data-driven algorithms

Xia et al. (2017) established a novel detection method based on the correlation coefficient to identify the short circuit faults in battery packs. The proposed method was executed without any hardware or analytical redundancy and worked effectively under the inconsistencies in SOC and SOH. Li and Wang (2018) adopted a novel correlation coefficient-based fault diagnosis approach for the lithium-ion battery pack in EVs. The proposed method not only amplified the voltage difference among batteries but also extended the fault memory by assigning a suitable size of moving windows. Kang et al. (2019) proposed a multi-fault diagnostic approach based on interleaved voltage measurement and recursive correlation coefficient to identify sensor faults, short circuit faults, and connection faults. The developed method detected the faults accurately with two different sensors and employed the enhanced correlation coefficient to address the cell inconsistencies and measurement error. Kong et al. (2018) suggested a quantitative diagnosis method to identify the micro-short circuit fault of lithium-ion batteries based on variation in remaining charging capacity. The performance of the proposed model was examined against different charging, aging conditions, and results were satisfactory in terms of computing micro-short circuit resistance and the leakage current with computational efforts. Xia and Mi (2016) introduced a fault-tolerant voltage measurement approach based on matrix interpretation for series battery packs. An electronic circuit was designed to separate sensor and cell faults by locating the abnormal signals. The experimental results proved the practicability and robustness of the proposed method concerning diagnostic accuracy, cost, and diagnosis confidence. Xia et al. (2016) concentrated to distinguish the faults between sensor and battery cells using an enhanced interleaved voltage measurement method. An improved circuit topology was designed and the model showed improvements in reducing noise levels compared to the basic interleaved topology.

3.3.4. Entropy algorithms

Wang et al. (2017b) presented Shannon entropy-based a novel fault diagnosis method to diagnose the voltage faults in battery systems. The proposed diagnosis method was evaluated under real-time voltage monitoring data and showed excellent forecast results in terms of the time and location of the voltage fault in battery packs. Liu et al. (2018) applied entropy theory to detect the voltage abnormality for a multiple-cell battery system. The entropy weight was introduced to further improve reliability by identifying the abnormal cells and predict faults. Li et al. (2020a) proposed a novel technique employing sample entropy and empirical mode decomposition (EMD) to detect faults in BMS under numerous operational settings. The EMD was used to extract the fault features experimentally using the voltage signal decomposition and noise interference cancellation. After, sample entropy detected the battery faults accurately based on extracted fault features. The real-world EV data was utilized to validate the performance of the proposed method under normal and sudden faults.

3.3.5. Other algorithms

Kim et al. (2018) introduced a real-time fault diagnosis strategy utilizing the outlier detection algorithm for multi-cell lithium-ion batteries. The Z-score standardized pre-processing method was used to detect the abnormalities in battery cells and identify faults in healthy battery cells, aged battery cells and shorted battery cells. Table 8 shows the evaluation of different kinds of faults analyzed by intelligent algorithms and controller schemes in BMS.

3.4. Thermal management system in BMS

The high temperature of the battery cell is a serious concern that needs to be addressed properly to improve EV performance. A thermal runaway of battery usually occurs when high temperatures activate the exothermic reactions which accelerate the temperature further and leads to more damaging reactions (Hannan et al., 2020). The impacts of thermal runway under different categories of lithium-ion batteries are shown in Fig. 8. It is observed that the electrolyte, cathode and solid electrolyte interface (SEI) layer start decomposing when the temperature increases beyond 90 °C, However, $LiFePO_4$ exhibits limited exothermic heat discharge, thus providing improved thermal stability in comparison to other lithium-ion batteries (Lu et al., 2013). Wang et al. (2016) found that the lifespan of the battery had become shortened by two months if

Table 7
Analysis of control strategies used for battery equalization in BMS.

Refs	Equalization method	Type of converter	Variables	Objectives	Strength	Weakness	Contributions	Research Gaps/Future Works
Wang et al. (2020a)	FLC	-Buck-Boost Converter	-Voltage -SOC	- To minimize equalization time. - To enhance efficiency.	-FLC has a high fault tolerance capability. - FLC has high equalization time and efficiency.	-FLC has poor flexibility. -FLC rule generation relies on knowledge and human expertise.	- Improves the equalization time and efficiency by 49% and 4.88%, respectively.	-The temperature impact and capacity degradation were not considered in SOC estimation. -The path optimization algorithm is required to reduce energy loss and increase equalization speed.
Ma et al. (2018)	FLC	-The two-stage bidirectional equalization circuit	-SOC	- To reduce SOC deviation. - To minimize equalization time.			-Minimizes SOC deviation and equalization duration by 18.5% and 23%, respectively.	- The proposed equalization method was not validated under aging impacts. - The equalization effect under the battery module can be studied.
Nguyen et al. (2014)	Neuro-fuzzy	-Flyback converter	-Current -Voltage	-To improve learning ability from various working conditions.	-This method can address nonlinear, complex, and dynamic systems. -It has an easy implementation process both in hardware and software.	-This method needs the influx of data to achieve high precision.	-Obtains a learning accuracy error of $1.8 \times e^{-5}$.	-The membership function of the input variables was not optimized.
McCurlie et al. (2017)	MPC	-Flyback converter	-SOC	-To enhance the efficiency of power electronics switching. -To extend battery life.	-This method is appropriate for time-varying, nonlinear, and multivariable processes. -It can intrinsically handle the constraints.	-This method is computationally demanding. -The solution is not guaranteed.	-Balancing terminates when SOC difference is 2%.	- The efficiency could be hampered due to the switching loss and conduction loss.
Liu et al. (2017a)	MPC	-Bi-directional Buck-Boost converter	-SOC -Energy	-To minimize energy dissipation during balancing.			-Achieves 93% efficiency in balancing.	- This research could be extended for a series-parallel combination of battery packs to balance the cell capacities.
Zhang et al. (2015)	GA	-Two-stage DC/DC converter	-SOC	-To improve energy efficiency. -To minimize equalization time.	-The concept of GA is easy to understand. -GA is inherently parallel, easily distributed.	-GA takes a long time to obtain near-optimal solutions. -GA is sensitive to input parameters.	-GA scheme reduces equalization time by 1002.5 s. -GA scheme improves energy efficiency by 93.1%.	- GA takes a substantial amount of computing time. -The superiority of GA over other optimization algorithms can be assessed.
Sun et al. (2015)	PSO	-Bidirectional DC/DC converter	-SOC	-To reduce battery pack imbalance and energy loss. -To improve battery pack performance.	-PSO requires fewer parameters to adjust. -PSO has a simple execution. -PSO has high efficiency and fast convergence speed.	-PSO can converge prematurely. -Hard to define the initial design parameters.	-Achieves the maximum SOC gap within 2% -Improves the capacity by 13.2%	- A detailed comparative analysis is required to prove the PSO effectiveness. -The battery cell SOC variation and temperature influences can be considered.
Liu et al. (2017b)	ACO	-Bidirectional DC/DC converter	-Voltage	- To reduce the energy loss -To achieve high balancing efficiency and high speed.	-ACO has guaranteed convergence. -ACO has the adaptability to changes in new solutions.	-ACO has a complex theoretical study. -The convergence duration of ACO is uncertain.	- Obtains total energy loss up to 1.53 in the entire balancing process.	- The formulation of multi-objective function and other key factors need to be considered to achieve better equalizer efficiency. -Comprehensive experimental verification is required.

the battery temperature rises 1 °C in the temperature range of 30 °C–40 °C. Therefore, the heat management in the battery system for EV applications is very crucial and further exploration is required to design a suitable cooling and heating system for BMS.

An effective thermal management system (TMS) is essential to maintain the battery temperature within the boundary toward the optimum performance of BMS (Zhang et al., 2018a). The operating temperature of the battery is controlled by the TMS so that each battery cell must be operated inside a threshold temperature range. When the temperature of the battery reaches outside the safe limit, TMS provides immediate response through the heating and cooling management and control system thus ensuring safe operation and protecting the battery from harmful occurrences (Jacob et al., 2019). Battery TMS either employs air as the heating/cooling ventilation or liquid for the cooling/heating insulation layer, as illustrated in Fig. 9. The air TMS covers passive cooling, passive cooling/heating, and active cooling/heating, as depicted in Fig. 9 (a) while liquid TMS includes passive cooling, active cooling, and active cooling and heating, as shown in Fig. 9 (b). The liquid TMS delivers better solutions with regard to heat capacity and thermal conductivity. An electronic control unit is used for the control operation of TMS (Hong et al., 2017).

3.4.1. Controller schemes of TMS based on cooling and heating

Hannan et al. (2019) proposed a robust control system for TMS using PSO based fuzzy logic controller (FLC). The effectiveness of fuzzy was compared with the PID controller with respect to overshoot, undershoot, and settling time. The proposed method improved the battery performance by controlling the temperature and consequently achieved overshoot and undershoot of 0.497% and 0.975% during the heating and cooling subroutine, respectively. Ma et al. (2014) designed a novel TMS based on FLC to handle the inconsistencies of series-connected lithium-ion batteries. The developed model increased the safety of the battery system effectively by

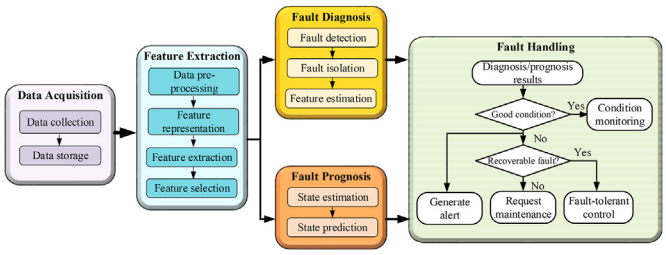


Fig. 7. General fault diagnosis framework for BMS (Niu, 2016).

regulating the temperature and balancing SOC during the charging and discharging processes. The FLC demonstrated strength in reducing SD and equalization time of SOC by 18.5% and 23% respectively. Umair Ali et al. (2018) demonstrated FLC based fast-charging approach and temperature control unit for a lithium-ion battery to protect the battery from overvoltage and overheating. Rahman et al. (2015) presented an advanced evaporative cooling (EC) TMS based on FLC to regulate the temperature of the battery in the limit of 20–40 °C during charging and discharging mode. The Proton Saga EV was employed to conduct the experiments and results were promising with regard to energy, saving 17.69% compared to air cooling battery thermal management. Xia et al. (2018b) presented FLC based TMS to control the battery temperature toward the enhancement of battery life, and performance. FLC can operate under various uncertainties, nonetheless, it lacks flexibility and needs detailed knowledge and experience to develop fuzzy rules. Afzal and Ramis (2020) introduced a hybrid multi-objective optimization method based on GA and FL to develop a thermal management system for a lithium-ion battery system. The maximum temperature, Nusselt number and friction coefficient were selected as the objective function. The results indicated that the proposed model achieved the desired temperature range with the improved Nusselt

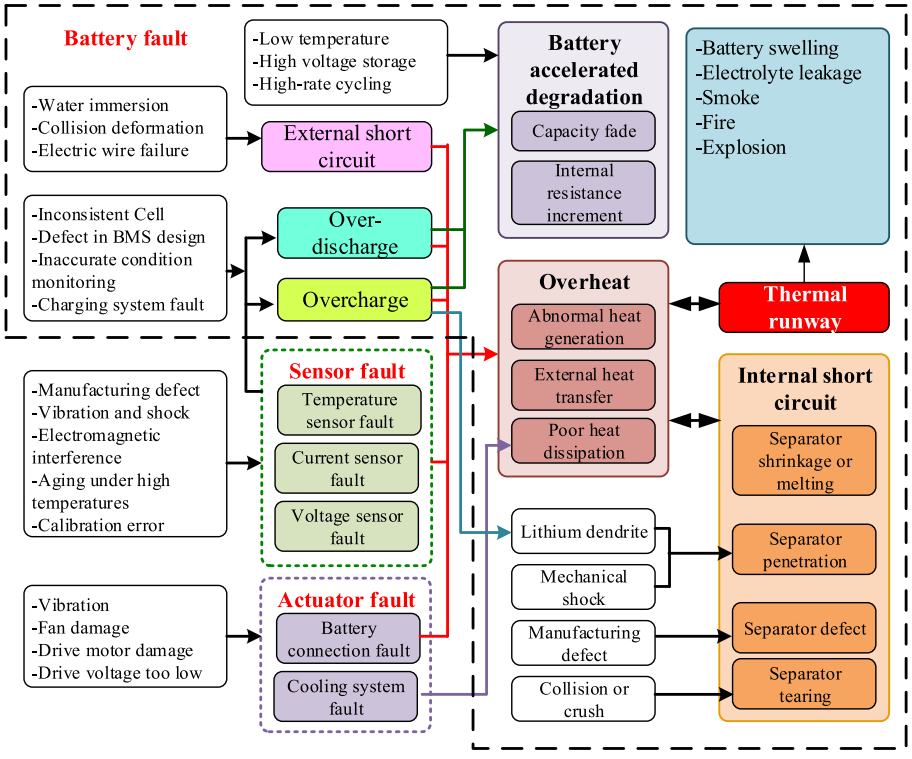


Fig. 6. The outline of various faults in the lithium-ion battery systems.

Table 8
Analysis of various fault detection methods based on intelligent algorithms and controller schemes in BMS.

Ref.	Methods	Type of faults	Faults characteristics	Benefits	Shortcomings	Contributions	Research Gaps/Future Works
Xiong et al. (2012)	Rule-based method	-Overcharge	- The rise of Temperature. -The drop-in temperature.	-Easy process and implementations.	-Difficult to estimate the suitable parameters in the rules	-Real-time execution -Performs early warning and failure detection.	-This method can be employed for series and parallel connected lithium-ion battery pack. - The proposed method can be tested with other types of lithium-ion batteries.
Muddappa and Anwar, (2014)	Fuzzy logic	-Aged battery -Overcharge -Over-discharge	-Derived from voltage, temperature, and SOC parameters.	-Easy to operate with uncertainty.	- Poor self-learning Skills. -High computational cost.	-The reliable and quick diagnosis method - Detects various faults including aged battery, overcharge, and over-discharge	-The experimental validation is required for real-time battery applications.
Yang et al. (2018)	RF	-Electrolyte leakage -ESC fault	-Maximum temperature rise. -Discharge capacity.	-Improved classification capability. -Fast execution process.	-Accuracy depends on a large number of training datasets. -Abundant fault data is not easily accessible.	-Diagnosis of the ESC faults and leakage faults accurately under several cases. -Low computational efforts.	-The accuracy and robustness of RF can be further checked with other regression methods such as SVM, RVM.
Zhao et al. (2017)	FNN	-Battery fault	- Abnormal changes in battery terminal voltages.	-Reliable and Accurate. -Not sensitive to uncertainty	-A large amount of training dataset is needed -Data abundance for training operation is not easily obtainable.	-Performs the fault diagnosis and prognosis with big data statistical regulation.	-Further research can be performed on battery fault taking place in winter.
Hong et al. (2019)	LSTM	-Battery fault	-Voltage abnormities.	-Powerful prediction ability.	-Hyperparameters adjustments is a laborious task. -Time-consuming training operation.	-Detects the advent of battery faults. -Mitigates runway risks.	-Performance of LSTM can be improved with parameter optimization. - The dataset and model need to be updated periodically to obtain better results.
Wang et al. (2017b)	Shanon Entropy	-Voltage fault during EV operation	-Irregular changes in battery voltages.	-Wide Range application. -It can predict fluctuations in the chaotic system.	-The entropy outcomes are affected by the computation window -It has a high computational burden.	-Assesses the time and location of the voltage fault accurately.	- Future work can be performed using a larger battery pack under a complex environment. -Data acquisition time interval could be expanded to 30 s in the future research
Kong et al. (2018)	Wavelet transform	-Short circuit faults	-Remaining charging capacity variation.	- It has low computational complexity. -It can map the signals more efficiently than the Fourier transform.	-Poor directional selectivity -unpredictable variation in output coefficients.	-Determines the micro-short circuit resistance and the leakage current with computational efforts.	-Further research could be extended using an aging impact on diagnostic results. -Experimental tests are required for validation under short circuit operation.
Kang et al. (2019)	Correlation coefficient	-Sensor faults -Short circuit faults -Connection faults	-Voltage inconsistencies.	-Easy to apply and understand. -Accurate and Robust.	-It has a complex matrix interpretation. -Difficult to formulate a mathematical correlation coefficient. -Time-consuming method.	-Robust to cell irregularities and measurement noise. -Detect the location and type of faults precisely.	-This topology can be redesigned with a lower number of voltage sensors without affecting diagnosis results.
Xia et al. (2016)	Sensor topology	-Sensor faults -Battery cell faults	-Voltage variations.	-It works efficiently without hardware or software redundancy.	-High mathematical computation. -Complex circuit design.	-It has a lower noise level. - It has broader application by removing the coprime constraint	-This research could be extended for multiple same subjects where the fault tolerance level is critical

number and lower friction coefficient. Although the integration of FLC with heuristic algorithms can deliver satisfactory outcomes, it has limitations in terms of fitness function selection and computational time. Lopez-Sanz et al. (2017) introduced a data-driven nonlinear model predictive (NMPC) control strategy to control the temperature of high voltage battery cells. The proposed method maintained healthier temperatures of the battery and reduced the cooling electrical consumption and temperature by around 38%. Altaf et al. (2016) built a robust electro thermal control using a data-driven model predictive control (MPC) method to prevent non-uniform aging as well as thermal and SOC imbalances. MPC can extend to time-varying, multivariable process and constraints; but it has weaknesses concerning model construction and guaranteed solutions. Table 9 shows the performance of different controller schemes for TMS of BMS.

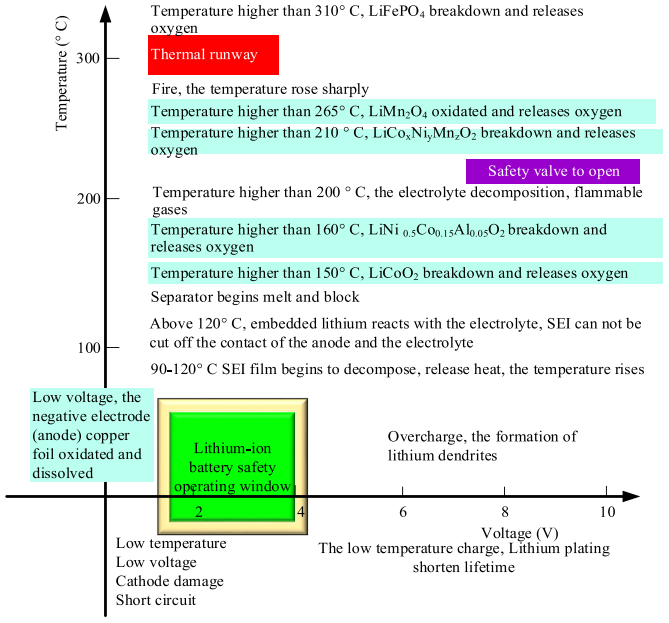


Fig. 8. Effects on high temperatures on different lithium-ion battery materials (Lu et al., 2013).

4. Key issues and challenges of BMS

The implementation of intelligent approaches employed in BMS for EV applications has become a major concern due to the algorithm complexity as well as various internal and external factors. The key issues and challenges are divided into three groups; algorithms issues, implementation issues and battery-related issues.

4.1. Algorithms issues

Although intelligent techniques have demonstrated promising solutions toward the development of an efficient BMS, they have some limitations. FNN-based approaches can achieve reasonable results; however, performance is constrained due to storage capacity and execution time. The regression and probabilistic approaches are robust against noise, uncertainty and data-overfitting issues; however, they fail to provide accurate solutions in high dimensional and non-linear systems. The time series-based methods can predict future predictions accurately; nevertheless, the selection of past information and feedback steps is difficult to determine. Although deep learning algorithms have provided excellent results in SOC, SOH, and RUL prediction, they need good quality and a vast amount of data as well as fast and expensive computing processors. The optimization strategies can be integrated with machine learning and deep learning algorithms; nonetheless, accuracy could deviate due to the local minima trap, poor search capability, and wrong parameter settings. The entropy algorithm can measure the complexity and variation in the measured signals, nevertheless, it has a shortcoming of the high computational process. MC algorithm is outstanding in probability and complex distributions, nonetheless it is expensive and needs heavy computation power. The statistical data-driven based algorithms can operate under noise and degradation function; however, they have a high mathematical computation process which leads to slow convergence. The fuzzy controller is reliable, customizable, efficient; however, it requires lots of data and needs human expertise. MPC has improved transient response and can control multiple variables within the boundary, but it suffers from the high computational load, high algorithm complexity, and the high number of control parameters.

4.2. Implementation issues

This section covers the various implementation barriers of intelligent approaches and control schemes of BMS in terms of data variety, data integrity, algorithm complexity and uncertainties.

4.2.1. Data abundance and data variety

The data abundance and data variety are the key issues in executing intelligent algorithms. The precision of intelligent methods relies on sufficient quantity and quality data.

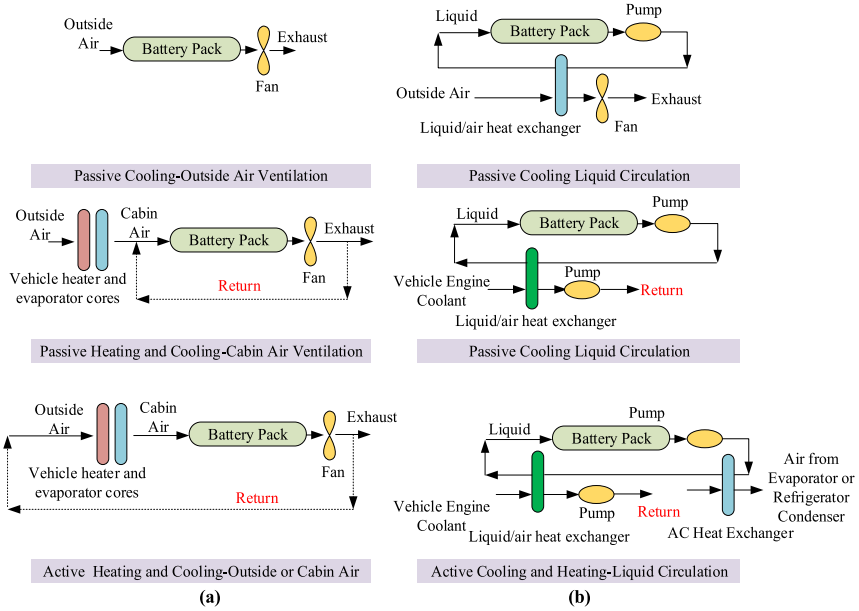


Fig. 9. Schematic diagram of battery TMS based on air and liquid (a) Air TMS, and (b) Liquid TMS (Hong et al., 2017).

Table 9
Literature survey of the controller schemes for TMS of BMS.

Intelligent methods	Ref.	Thermal Type	Target	Pros	Cons	Achievements	Research Gaps/ Future Works
FLC	Hannan et al. (2019)	Heating and Cooling	-To control temperature during the high rate of charging and discharging process -To minimize irreversible capacity loss and degradation of battery performance.	-The membership functions (MFs) of FLC are optimized using PSO. -Can maintain a consistent temperature among the battery pack.	- The method is validated with only 10 series-connected battery cells. - The fitness function of optimized MFs is not assessed in the convergence curve.	-Obtains an overshoot of 0.497% during the heating subroutine. -Achieves undershoot of 0.975% during the cooling subroutine. -Attains settling time of 32 min 13 s and 28 min 46 s during heating and heating.	-This topology can be validated using a prototype for a lithium-ion battery storage system.
FLC	Ma et al. (2014)	Cooling	-To regulate the temperature. -To balance SOC.	- Can balance SOC and temperature among cells under both charging and discharging operation.	- The results are verified through simulation study only.	-Decreases the equalization time by 23% -Reduces the standard deviation of the final SOC by 18.5%.	-Appropriate hardware validation is required to demonstrate the suitability of the proposed method.
FLC	Umair Ali et al. (2018)	Heating and Cooling	-To develop an efficient, real-time, fast-charging approach under temperature feedback.	-Protects the battery from overvoltage and overheating. -Fast-charging operation. -Keeps the battery within SOC limits.	-Unsatisfactory performance in over-cooling control. - Improves the SOC and charging rate while avoiding carbon emission cost.	-Takes 9.76% less time in complete battery charging compared to the conventional method. - Charge capacity can reach to 99.26% SOC without causing any performance degradation.	-The MFs of FLC can be tuned properly for better charging efficiency. - The methodology can be implemented using multi-cell battery cells.
FLC	Rahman et al. (2015)	Cooling	- To increase the battery life span	- Can control the temperature of the battery pack within 20–40 °C during both charging and discharging mode.	-MFs of FLC are not optimized. -Generation of fuzzy rules is a tedious task.	-Saves energy by 17.69% compared to air cooling battery thermal management.	-Performance investigation at higher temperatures can be carried out. -The proposed model can be tested under different drive cycles.
MPC	Lopez-Sanz et al. (2017)	Cooling	-To control the temperature of high voltage battery cells.	-Fast response. -Ability to reduce thermal stress and energy consumption of components.	- SOC, level of charge and life cycle of batteries are not considered. - The battery pack under low voltage is not examined.	-Minimizes the high temperature efficiently. -Keeps the battery temperature at a safer range. - Decreases cooling energy by 5%.	-The costs, durability, complexity and performance of the proposed method in TMS need to be investigated.
MPC	Altaf et al. (2016)	Cooling	-To prevent non-uniform aging. -To reduce thermal and SOC imbalances.	-Can control the temperature with incomplete driving information -Both voltage control and balancing control can be achieved.	- The balancing between SOC and over-cooling is not carried out. -The controller is suitable for hot environment only.	-Maintains the balance between over-heating and SOC. - Has negligible voltage error (<3.2%) under model uncertainties and cell temperature variation.	- Experimental verification is required for large battery packs - Further studies can be conducted on battery module size and electronic component losses.
Observer-based controller	He and Ma, (2015)	Cooling	- To minimize the temperature and maintain the colling flow among cells.	-Ability to track the cell core temperature accurately.	-Suffers from the slow response in comparison to fuzzy.	-Non-uniformity in temperature among cells is decreased from 4.2 to 1.0 °C -Cooling flow consumption is reduced by 38%.	-A comprehensive experimental demonstration is needed to verify the controller's effectiveness.

Nevertheless, the collection of large amounts of data with enough variety is time consuming and laborious task. In general, the data is collected through experiments at a sampling frequency of 1 Hz. The data length among EV drive cycles varies with different shapes of voltage and current values (Hannan et al., 2020b). For instance, the one complete cycle for FUDS, DST, BJDST and US06 is computed to be 1372 s, 360 s, 916 s and 600 s, respectively (Zheng et al., 2016). Since the training accuracy of intelligent algorithms needs a large pool of data, thus multiple repetitions of EV drive cycles are employed for data preparation (Yang et al., 2020). A large amount of data can support to achieve better solutions, however, it may result in computational burden and extend the training time leading to over-fitting issues (Najafabadi et al., 2015). Thus, special attention is required on data abundance and data variety issues.

4.2.2. Data integrity

Data integrity is another barrier in implementing intelligent methods in a real-world environment. A few of the recognized automotive research groups (CALCE, 2017; Kollmeyer, 2018; Kollmeyer et al., 2020; Y. Wang et al., 2017a) have released the high-quality publicly available battery dataset which contains a fixed pattern of charge-discharge current ensuring the various protocols of EV drive cycles. The experiments to collect the various EV drive cycle data are carried out in the laboratory environment using the recommended charge/discharge current rates and temperature conditions. The obtained EV drive cycles are the simulated data and their current and voltage profiles would not be as similar as data under the real-world environment. Hence, the verification of intelligent algorithms under practical cases needs further investigation.

4.2.3. Suitable structure and hyperparameter adjustment

The effectiveness of intelligent algorithms relies heavily on input features, network architecture, training algorithms, activation functions and hyperparameter adjustment. So far, lots of efforts have been made by different researchers to design a suitable structure for intelligent algorithms. Accordingly, the several hyperparameters are adjusted to develop an improved framework for intelligent algorithms including HLs, HNs, learning rate, weight, bias, membership function, timestep, iterations, batch size, sampling interval etc. Besides, various training and activation functions are employed to execute intelligent methods including Sigmoid, Tangent, Gaussian, Radial basis kernel, Gradient descent, Levenberg-Marquardt, Logistic regression, backpropagation etc. The appropriate selection of hyperparameters and functions decreases the computational complexity resulting in the reduction in data underfitting and overfitting issues. In the majority cases, the intelligent models are configured by tuning the hyperparameters and functions based on trial and error method, which leads to loss of human energy and substantial time. Therefore, suitable design and hyperparameter adjustment of intelligent algorithms are the key issues to be explored.

4.2.4. Optimization integration issues

The optimization integration into intelligent methods and control schemes is a challenging and laborious task. Numerous optimization methods could be combined with intelligent approaches. However, the performance of optimization varies from one algorithm to another in terms of execution time and convergence speed. Besides, optimization algorithms are structured using a lot of parameters and operational stages. The parameter initialization and operational loop execution in optimization need depth knowledge and human expertise. Although the inclusion of optimizations into intelligent methods in BMS has demonstrated substantial contributions in terms of accuracy, robustness and efficient prediction, they have some negatives concerning computational complexity and long training interval. The integration of optimization may lead to unsatisfactory outcomes if the convergence, search capability, dimension settings are not addressed properly. Therefore, the selection of suitable optimization technique is vital to attain acceptable results in prediction and control operation in BMS.

4.2.5. Algorithm verification uncertainties

The performance of intelligent algorithms could diverge due to the presence of uncertainties such as erroneous initial state value, equipment error, electromagnetic interference (EMI) noise, EV drive cycle patterns, temperatures and many unknown factors. In practical cases, the initial SOC may not be 100%. Hence, the accuracy of the state estimation in BMS can be evaluated under different initial SOC values. For instance, the performance of GRU was checked under 100%, 80%, 60%, 40% and 20% initial SOC (Yang et al., 2019a). Moreover, the battery measurements may be affected by low sensor precision, EMI and noise effects (Kang et al., 2014). Thus, different types of noises can be added to check the performance accuracy of intelligent algorithms. For example, Cui et al. (2018) verified the adaptability and robustness of WNN based SOC estimation approach with the presence of bias noises and random noises. Furthermore, numerous EV drive cycles are employed to assess the adaptability and robustness of the WNN algorithm. It is reported that the accuracy of SOC was varying due to different patterns of current and voltage measurements in EV drive cycle. Another study by Li et al. (2019a) validated the GRU algorithm for SOC estimation under various EV drive cycles, indicating MAE of 2.12%, 1.13%, 0.71% and 1.01% in HWFET, LA92, UDDS and US06, respectively. Moreover, the temperature has a huge impact on battery state estimation. Leng et al. (2015) found that the capacity of lithium-ion battery started decreasing when the temperature increased above 45 °C. Chemali et al. (2018) applied the DFNN in battery SOC estimation of BMS under wide-ranging temperatures at −20 °C, −10 °C, 0 °C, 10 °C, and 25 °C. The results illustrated the varying MAE between 1.4% and 2.2% in HWFET drive cycle. Therefore, the verification of intelligent algorithms in BMS under diverse EV drive cycles and temperatures is essential toward accurate and robust state estimation in BMS.

4.3. Battery related issues

The accuracy of state estimation in BMS could be deviated due to the various battery related issues such as battery chemistry, aging cycle, thermal runway, capacity fading, charging and balancing problems. The accuracy of battery state estimation under different battery chemistries could be different even if the same intelligent algorithm is applied. For instance, Lithium Iron Phosphate (LiFP) batteries exhibit cost-effectiveness and outstanding safety features while Lithium Titanate (LiTO) batteries offer a long lifespan (Zhang et al., 2018c). Chaoui and Ibe-Ekeocha (2017) assessed the performance of recurrent NARX algorithm in SOC estimation and the results indicated RMSE of 0.53% and 0.7% in LiFP and LiTO batteries, respectively.

Battery aging is another cause to degrade the accuracy of intelligent algorithms and controller strategies in BMS. Leng et al. (2015) reported that the rise of aging cycles resulted in various issues such as degradation of internal resistance, capacitance, structure variation of the anode, cathode material and growth of solid electrolyte interphase (SEI) thickness, leading to a variation in battery SOC, SOH and RUL prediction in BMS. For instance, Li et al. (2016) validated the GPM method for RUL prediction using NASA PCoE data and computed RMSE of 1.58% and 1.3% in 60 and 80 cycles, respectively. Another study by Li et al. (2014b) predicted

SOH using RVM method and reported RMSE of 0.17%, 0.24%, 0.36% and 0.22% in 500, 600, 700 and 800 aging cycles, respectively.

The performance of intelligent algorithms and control schemes in EVs could hamper due to the high battery temperature. The thermal runaway is a serious concern that can be occurred by mechanical and electrical abuse resulting in lithium plating, overcharge, internal short circuit, exothermic reactions and collision problems (Zhang et al., 2018a). The growth of SOC and high aging cycles are the root cause of thermal runaway (Galushkin et al., 2018). The negative electrode, SEI layer and electrolyte begin to decompose when the temperature is over 90 °C (Lipu et al., 2020a). Hence, various studies are carried out on thermal runway and modeling including analytical model (An et al., 2019), 2D model (Bugryniec et al., 2020), 3D model (Feng et al., 2016) and kinetics analysis (Ren et al., 2018).

The battery capacity and power fading are other responsible factors in deteriorating the performance of intelligent algorithms and controller schemes in BMS. The capacity loss of the battery starts when the transformation of the battery active material begins under any discharge rates (Ramadass et al., 2002). In contrast, the power rate capability is noticed when the rise of battery impedance and decrease in operating voltage are occurred (Panchal et al., 2017). A study by (Amine et al., 2005) reported that the capacity of the battery is reduced when the temperature rises above 45 °C under high aging cycles. Another study mentioned that, battery capacity loss takes place when the voltage climbs up above the maximum value, as suggested by the battery manufacturer (Takei et al., 2001).

The charging method of lithium-ion batteries is crucial to obtain a good balance among heat, efficiency, lifespan and health. A slow charging method could hinder the widespread acceptance of EVs while the fast-changing approach generates heat and reduces lifespan (Yin et al., 2019). On the other hand, charge imbalance among batteries is noted due to the variation in physical property, degradation in battery aging and capacity, material defects, manufacturing technology and tolerances (Diao et al., 2019). The charge balancing issues of the battery results in divergence in battery state estimation as well as affect the safety mechanism of BMS (Turksoy et al., 2020). Hence, an effective charge balancing approach is required to elevate the performance of intelligent algorithms.

Aside from the abovementioned issues, the battery state estimation, temperature control and fault diagnosis in BMS can be influenced by the battery model, battery self-discharge, hysteresis, and charge/discharge current rate.

5. Future outlook and directions for BMS

Based on the comprehensive study and analysis of available literature of intelligent approaches and various control schemes of BMS toward state estimation, thermal management, fault diagnosis and battery balancing, the specific opportunities for future works are suggested which are discussed in the following sections.

5.1. Development of advanced sensing equipment

The monitoring of battery SOC, SOE, SOH, and RUL is carried out using different battery testing systems (BTSs) such as Digatron (Chemali et al., 2017), Arbin BT 2000 (Sahinoglu et al., 2018), DAQ (Chaoui et al., 2017) and NEWARE BTS 4000 (Wang et al., 2014b). Nevertheless, the measurements by BTS may not be accurate due to EMI, noise impacts, and equipment error (B. Xia et al., 2018b). In addition, the majority of state estimation intelligent algorithms are validated under laboratory environments which may not achieve the desired outcomes in real applications due to a large amount of noise caused by EMI and sensor flaws. Thus, the advanced sensing technology-based BTS is required to reduce the influence of the measurement noise and accordingly elevate the estimation accuracy. Various methods are introduced to reduce noise impacts including wavelength transform, Butterworth filter, bias compensating method, adaptive forgetting variant and recursive total least squares method. Wei et al. (2021) integrated bilinear principle and instrumental variable (IV) to evaluate the impact of measurement noises on model parameters and SOC estimation. The IV estimator was employed to co-estimate the model parameters and noise variances. A two-stage least squares algorithm was utilized to solve the co-estimation problem. The Luenberger observer was combined with the parameterization method to determine SOC online. The simulations and experiments results confirmed the model superiority with high noise immunity. Another study by (Wei et al., 2020) concentrated on the noise-immune and unbiased model parameter determination for lithium-ion batteries. A novel two-step method integrating variable projection algorithm (VPA) and least squares (LS) was employed to co-estimate the noise variances and unbiased model parameters. The computational efficiency of the proposed model was further enhanced using a numerical solver. Both simulations and experiments demonstrated excellent outcomes in term of parameterization accuracy and noise tolerance.

5.2. State monitoring for the battery pack in EVs

In the majority studies, the single lithium-ion battery is considered for SOC, SOE, SOH, and RUL prediction as well as fault diagnosis and TMS. However, EV is designed using a battery pack consisting of hundreds of lithium-ion battery cells. The accuracy of a single battery would not be as same as the accuracy of the battery pack due to the unbalancing issues that occur after repeated charge/discharge cycles. Although various controllers have been developed to address the above issues, further exploration is required in improving their performance with regard to cost, size, power loss, efficiency, execution, equalization time, and voltage/current stress. Awadallah and Venkatesh (2016) proposed an adaptive neuro-fuzzy inference system (ANFIS) algorithm to compute SOC for a lithium-ion battery pack consisting of 36 cells connected through a series-parallel combination. The battery SOC was evaluated at the cell level and then combined to calculate SOC for the battery pack. Sun et al. (2016) proposed RBFNN based SOC algorithm framework to determine SOC for a battery pack with twelve lithium-ion polymer batteries connected in series. Deng et al. (2020) introduced a data-driven technique for SOC estimation of lithium-ion battery pack addressing the cell inconsistencies and data complexity through correlation and principal component analysis.

5.3. Algorithm hybridization

Presently, the hybridization of intelligent algorithms has illustrated superiority over a single intelligent algorithm in improving the prediction and control performance. Generally, hybridization is formed by combining intelligent methods with the optimization technique. For instance, Hannan et al. (2020b) combined recurrent NARX algorithms with LSA to enhance the accuracy of SOC estimation. Chang (2013) applied GA and orthogonal least-squares algorithms to optimize the RBFNN model toward accurate and robust SOC estimation. PSO was integrated with BPNN (Zhang et al., 2018e) and WNN (Xia et al., 2018a), respectively to optimize the hyperparameters and improve SOC estimation accuracy. Nevertheless, the inappropriate combination of hybridization contributes to complex configuration and hence leads to mathematical

complexity and undesirable outcomes. Therefore, further studies are required and practicability issues need to be considered to develop an efficient hybrid algorithm.

5.4. Training performance enhancement

One of the major challenges to execute the intelligent methods is the training accuracy and duration specifically for the deep learning algorithm. Chemali et al. (2018) carried out the training operation of deep learning algorithms offline with thousands of battery data in around 50–60 h. In another study (Chemali et al., 2017), the computational time to train the battery data was estimated at around 4–9 h. Yang et al. (2019a) evaluated the training performance of the GRU network in SOC estimation under different epoch numbers and results indicated training time of 10.03 and 79.14 min in 1000 and 7000 epochs, respectively. The training time could be extended if the intelligent algorithm validation is conducted under more HLs, HNs and different temperatures. Therefore, further attention is essential to the selection of suitable activation functions, training algorithms, and hyperparameters to expedite the training process. In recent times, deep learning-based SOC estimation has received wide attention due to the high configuration processor with advanced graphics processing units (GPUs) and matured frameworks such as TensorFlow which has helped to reduce the training time (Sahinoglu et al., 2018). For instance, Song et al. (2019) used two GeForce GTX 1080 Ti GPUs to train the CNN-LSTM framework for SOC estimation of lithium-ion batteries. Li et al. (2019a) applied NVIDIA GeForce GTX 1070Ti GPU and 16 GB RAM to execute the training operation of GRU in SOC estimation.

5.5. Real-time embedded prototype design

To date, different experimental tests have been performed to validate the effectiveness of intelligent approaches toward state estimation, thermal management, and fault diagnosis. However, the implementation of intelligent methods in real-time BMS with small memory units and low computational cost has not been explored comprehensively. Hence, further investigation is needed to develop an embedded prototype system for real-time state estimation and control operation of BMS. There have been a few prominent researches on real-time state estimation in BMS using intelligent algorithms. Chen et al. (2019) validated the FNN based SOC estimation method on the hardware-in-the Loop (HIL) experimental platform in real-time. The HIL test bench was designed using a DC source, a current senor, battery monitoring unit, a host computer, a battery control unit and a CAN analyzer. The SOC error and capacity error of the proposed method in HIL test was obtained to be 2% and 19.7%, respectively. Singh et al. (2020) developed a HIL experimental setup to validate the ANFIS based SOC estimation in real-time using dSpace MicroLabBox hardware controller. The HIL results were very close to the simulated results which demonstrated the suitability of the proposed model in real-time EV applications.

5.6. Integrated with big data platform

The accuracy and robustness of intelligent algorithms of BMS under real-world environments can be further enhanced through cloud storage, cloud computing and big data platforms. The big data technology helps in executing intelligent methods with a large memory device, computation and analysis. The voltage, current, temperature, and other information of EVs are continuously transferred to the big data platform and accordingly, intelligent approaches can be trained under real-time tests and deliver more precise outcomes. The battery state estimation such as SOC, SOE, SOH, and RUL prediction along with thermal runway and fault diagnosis throughout the entire life can be monitored and saved in the cloud storage. Subsequently, the battery monitoring and control center will pre-process the data, carry out the analysis and deliver effective decisions for future performance improvement. Haldar et al. (2020) focused on real-time battery monitoring and management in EVs based on the internet of things (IOT). The authors developed a real-time wireless battery management system to monitor the battery discharging behavior and health status in EVs. Sivaraman and Sharmeela (2020) developed an IOT based BMS to monitor SOC, SOH as well as control the charge imbalance and fault diagnosis. Kim et al. (2018b) introduced a cloud-based battery monitoring system for the lithium-ion battery storage system. The proposed system was validated by cyber-physical testbench using Raspberry Pi3 IOT boards. The results illustrated that SOC and capacity were monitored accurately by the condition monitoring algorithm while the fault diagnosis was assessed by the data mining algorithm.

5.7. Multiscale and co-estimation process

The battery state estimation using multi-scale and co-estimations plays a vital role in improving the efficiency and reducing the computation of BMS. For example, battery SOC and SOE vary quickly with current fluctuation while the SOH and RUL change with a capacity variation. Thus, different time scales are needed to monitor the different states of the battery. Besides, the majority of the literature has suggested the methodology for single state estimation. The co-estimation or joint estimation combing two or three states has been very limited so far. A few investigations have been performed to shed some light on utilizing the intelligent algorithms for joint estimation. Hence, further studies are required on the co-estimation approach under different spatial and temporal scales. Chaoui et al. (2017) proposed an intelligent algorithm for two states (SOC and SOH) joint estimation for lithium-ion batteries. The battery SOC was estimated using the TDNN algorithm while the SOH was assessed using the normalized data in SOC learning. Another work by (Chaoui and Ibe-Ekeocha, 2017) presented RNN based co-estimation method for SOC and SOH estimation in lithium-ion batteries. SOH was predicted using the present and past values of current and voltage in SOC learning along with four milestones of battery lifespan under 100, 200, 1600 and 1800 cycles. The future outlook and opportunities for advanced BMS are shown in Fig. 10.

6. Conclusion

In recent decades, numerous analysis and evolution have been carried out on intelligent algorithms and control strategies of BMS in EVs due to the big data availability, powerful computation processor, and high capacity data storage devices. As a first contribution, this review explores the current progress of intelligent algorithms toward SOC, SOE, SOH, and RUL prediction, concentrating on structure, input features, pros, cons, and estimation error. The review has revealed that the intelligent algorithms have illustrated better performance in battery state estimation in terms of accuracy, adaptability, robustness, and efficiency. As a second contribution, this survey examines the various control techniques in temperature control, fault identification and prognosis and battery charge equalization, given importance on type, objectives, outcomes, advantages, disadvantages and research limitations. It is found that FLC and MPC can map a time-varying and complex battery system under constraints and uncertainties. As a third contribution, the review highlights the key concerns of intelligent algorithms and controller executions in BMS including data variety,

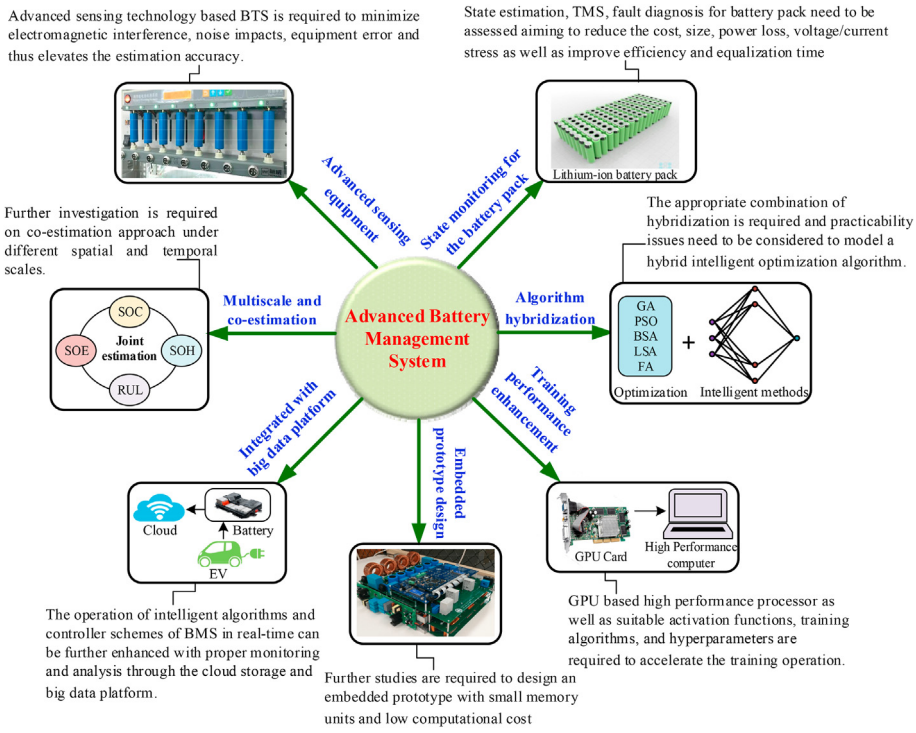


Fig. 10. Future trends for advanced BMS in EVs.

appropriate structure, selection of input features, hyperparameter optimization, uncertainties, computation complexity. As a fourth contribution, the review discusses various opportunities for future research works toward further technological development of intelligent approaches and control schemes of BMS in EV applications. This review proposes the future trends of BMS emphasizing advanced sensing equipment, embedded system, high-performance processor, hybridizing algorithms, multi-state estimation, model validation using battery module/pack and cloud computing technology.

In summary, the critical analysis and key information attained from this review would be beneficial for the automobile engineers as well as the EV industry in developing and executing advanced BMS for EV applications. Thus, further research works on BMS by intelligent algorithms and controller schemes will not only elevate the battery performance, lifespan but also ensure safe, reliable operation of EV resulting in substantial growth of the battery and EV market. The expansion of the EV and related battery market can help to achieve a pathway for sustainable development goals relating to clean energy, emission reduction, employment opportunities and economic development. Nevertheless, several shortcomings could hamper the wide adoption of EVs such as battery price and performance degradation, low driving mileage, poor vehicle features and ineffective marketing policies. Therefore, a long-term future initiative is necessary to enhance the EV performance with regard to accurate battery monitoring and controlling schemes development, investments, incentives, effective policies, carbon emission reduction, global collaboration and sustainable development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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